

~~From Public Data to a Coupled Data-Center Energy-Water Model~~ From Public Data to Network-Based Data-Center Planning

~~Meta Prineville as an Empirical Test Bed~~
~~Prineville Site Test Bed + M100 External Facility-Model Benchmark~~
Prineville Site Test Bed + External Facility-Model Validation
Facility Energy, Water Sources, Groundwater Response, and Decision Validation

~~Prineville Public-Data Baseline v1~~
Prineville Public-Data Baseline v2 — City utility-meter integration

Revision markup: green = prior external-validation revision; purple = September 3, 2026 project-status synchronization; blue-strikeout = superseded wording.

Short TODOs mark remaining identification or implementation tasks; they are not established results.

~~Current public main reports Prineville Public-Data Baseline v2. The generated Prineville report lists 50 sources, 82 quantities, and 26 registered methods/models; the validated City-integration run reported 126 package tests PASS. Exact snapshot provenance should be taken from git history and the generated registries rather than from a hard-coded hash in this glossary. This refresh is pinned to public main at 08b0c112d8883eb4d23bb05d0d40e1a42c863593. Deterministic reporting regenerated the authoritative 50-source, 82-quantity, and 26-method/model registries byte-for-byte; external validation streams remain outside the Prineville site-source count.~~

1 Big picture: what are we ultimately trying to model?

The long-run research objective is to connect a data center’s operation to the physical systems that bear its environmental impacts:

workload → IT power / heat → facility cooling and power → {grid, direct water} → {emissions /
generator water, water-system / groundwater state} → operations, siting, and policy.

The literature contains strong models for many individual links, but the links are usually studied separately. Operational papers often treat carbon or water intensity as exogenous signals; siting models often use static regional water coefficients; groundwater models take pumping schedules as given. The opportunity is to connect these layers while keeping the physical and accounting boundaries explicit.

The flagship validation is consequently a planning comparison rather than a facility-fit competition:

M_0 : static PSCC-style environmental coefficients,
 M_1 : validated facility/source/groundwater consequences.

The central question is whether the richer physical representation changes siting, technology/source choice, groundwater constraints, equity outcomes, or regret.

~~Prineville is the empirical test bed.~~ The present project asks how much of this full chain can be populated, reconstructed, and validated from public data for one real hyperscale campus before moving to coupled optimization.

Previous framing: Prineville site evidence + M100 as the only external facility benchmark.

~~Prineville remains the site-specific test bed; the internal facility model is now validated through a complementary external stack rather than a single site.~~

M100 provides measured facility-structure evidence; Lei-Masanet provides climate-technology PUE/WUE physics and archetypes; Frontier provides independent measured power/thermal evidence.

These sources constrain model form, while Prineville remains the site-specific electricity/water/grid/groundwater case.

Prineville remains the integrated site-specific test bed, while external experiments constrain model structure rather than supply a universal quantitative facility model. M100 is a frozen structural benchmark; Frontier provides closed independent evidence that IT load affects accessory power; ESIF adds measured evidence that facility overhead can change across architecture/operating regimes; Lei-Masanet remains useful for climate-technology and conditioning-water structure but its final quantitative adapter promotion is blocked; and Forest City v3 supports qualitative physical transportability while explicitly not validating numerical transfer. The facility layer is therefore substantially de-risked structurally, but site/archetype coefficients remain uncertain.

One view of the full model and current Prineville status

Table 1: One view of the full model and current Prineville status.
Statuses apply only at the stated physical/accounting boundary.

Layer	Main quantity	Current status	What that means in Prineville
Workload	jobs / utilization x_t, u_t	SCENARIO	No public Meta workload telemetry; current workload paths are synthetic.
IT power	P_t^{IT}	FITTED/ SCENARIO	Latent hourly shape; constrained indirectly by annual facility electricity.
Modern AI compute power	$P_{w,N,t}^{compute}$	frozen bounded external evidence	Controlled CPU/H100 compute-component energy is workload/mode/scale conditioned; full-node AC, other-node power, production mix, and the node-to-facility-IT bridge remain incomplete.
External facility structure	$P_t^{IT}, P_t^{cool}, P_t^{fac}$	EXTERNAL BENCHMARK	M100 Apr-Dec 2021 telemetry validates model structure and boundaries, not Prineville/generic coefficients.
Facility energy-water archetypes	$PUE_k(\mathbf{w}), WUE_k(\mathbf{w})$	external benchmark / validation in progress	Lei-Masanet first audit/reproduction identifies an IT-normalized climate-technology-intensity-model; publication-scale annual reproduction and RNG gates are currently running.
Facility energy-water archetype structure	$PUE_k(\mathbf{w}), WUE_k(\mathbf{w}), W^{cond}$	external benchmark; PARTIAL / closed	Lei-Masanet strongly supports climate-technology-intensity-structure and conditioning-water components, but the final reproducibility closure is PARTIAL and quantitative adapter promotion is blocked.

Layer	Main quantity	Current status	What that means in Prineville
Facility overhead regime/state	$P_t^{overhead}, A_t^{arch}, S_t^{op}$	external benchmark; PARTIAL / closed	ESIF shows that a stationary IT+weather model is insufficient for total auxiliary power across a late operating/infrastructure regime shift; use structure, not ESIF coefficients.
Facility electricity	E_y^{fac}	REPORTED	Annual Meta campus electricity is high-confidence ground truth.
Cooling	Q_t^{cool}, P_t^{cool}	DERIVED	Simplified gray-box physics driven by IT load, weather, and assumed parameters.
Weather	T, T_d, T_{wb}, p	MEASURED / DERIVED	NOAA KS39/KRDM observations; wet bulb-derived psychrometrically. Canonical hierarchy is KS39 when QC-usable, KRDM observational fallback, then bias-adjusted KBDN as tertiary observational fallback; RH/wet bulb are recomputed after the final mixed-source selection.
Campus water	W_y^{Meta}	REPORTED	Annual Meta withdrawal is reported for 2014–2024. It is withdrawal, not consumption.
Subannual campus water	W_T	scenario / weak prediction	No monthly campus meter; current allocations/predictions are not telemetry.
City customer-service water	$W_m^{City, service}$	REPORTED (observed meter report)	Monthly Facebook Data Center WATER-COMM + ADD’L WATER customer-service component is observed for 2012–2026 (2026 through July); it is not total Meta campus withdrawal.
All-source monthly campus withdrawal	$W_m^{Meta, total}$	NOT IDENTIFIED	Bulk/hydrant, SWR, WELL-METER-FOR-SEW, Trailer City, Warehouse, direct POD, reuse/return/storage, and parent/submeter semantics prevent a closed monthly campus balance.
Regional grid demand	D_t^{PACW}	MEASURED / PROXY	EIA-930 when available; FERC-constrained proxy for earlier years.
Grid carbon	CI_t	DERIVED / PROXY	Regional average/accounting context; not Meta-specific marginal emissions.
Generator water/emissions	$W_{g,t}, CO_{2,g,t}$	REPORTED / DERIVED	Strong plant context; no claim that particular plants physically serve Meta.
Water-system pumping	$Q_{i,t}$	REPORTED	OWRD municipal/direct-POD pumping at specific reporting boundaries.
GWIS groundwater level	$BLS_{i,t}$ (and reported AMSL)	MEASURED, qualified	QC-Individual-well observations; measurement-condition uncertainty remains.
Within-well relative head	$\tilde{h}_{i,t} = -(BLS_{i,t} - \overline{BLS}_i)$	DERIVED	Physically signed within-well anomaly used for the planned empirical benchmark; not a spatial groundwater-state field.
Groundwater response / state model	pumping $\rightarrow \tilde{h}$; future $h_{n,t}$	NOT IDENTIFIED	Next experiment tests whether antecedent pumping predicts within-well anomalies; a common-datum/network state is not identified.
Optimization	workload/siting/source choice	NOT IDENTIFIED	Future layer; response functions and costs are not yet sufficiently identified.

2 Evidence stack: who provides the data, and what boundary do they observe?

The strongest feature of the pipeline is that it does *not* treat all public data as if they measured the same thing. Different authorities observe different physical or accounting boundaries.

Table 2: Authority/evidence stack. Each row identifies the physical or accounting boundary contributed by that source; rows are not interchangeable measurements.

Authority / entity	What it is	What we use	Boundary / why it matters
Meta	Data-center operator	Annual Prineville electricity, withdrawal, location-based Scope 2, selected site/design disclosures	Campus-level ground truth; no public hourly IT or customer-meter telemetry. High-confidence annual reporting boundary; no public hourly IT telemetry. Exact PRN-only versus PRN+CCO inclusion by reporting vintage remains unresolved.
CINECA / M100 ExaData [1]	External measured Tier-0 facility benchmark	Facility/IT power, cooling components/state, node power, liquid thermal telemetry, weather	Validates generic facility-model structure and boundaries; not Prineville telemetry and not a source of generic coefficients.
Lei-Masanet [10] + public code	Climate/technology facility archetype model	Joint PUE/WUE and explicit conditioning-water components across published cooling cases	Used as an intensity/archetype backbone; annual reproduction is gated before project translation. It does not identify source withdrawal or groundwater pumping.
Lei-Masanet [10] + public code	Climate/technology facility archetype evidence	PUE/WUE intensity structure and conditioning-water components	Final reproducibility disposition PARTIAL; adapter promotion NO/blocked. Retain qualitative/structural archetype evidence only; no source withdrawal or groundwater inference.
Frontier [3]	Independent measured liquid-cooled HPC benchmark	Compute/accessory power, coolant temperatures/flow, waste-heat fields	Tests IT-load/accessory power and thermal structure on a different facility; not a water-withdrawal/WUE source. Gap-safe time-grid QC is being finalized.
Frontier [3]	Independent measured liquid-cooled HPC benchmark	Compute/accessory power and coolant/thermal fields	Corrected expected-time-grid audit is closed (coverage about 93.4%); the F1-vs-F0 load-dependence conclusion is unchanged. Thermal $\rho c_p \dot{V} \Delta T$ remains published-formula reproduction, not independent conservation validation.
NLR/ESIF HPC Facility PUE Data [4]	Independent measured facility-overhead benchmark	Measured IT, cooling, HVAC, pumps, plug/light, source PUE, weather	Electrical layer closed PARTIAL. Stationary IT+weather fails for total auxiliary power across the 2024 regime shift; architecture/operating state must be allowed, without coefficient transfer.
NLR / modern AI IT-power layer	External compute-component evidence	Controlled CPU/H100 workload-, mode-, scale-, and temporal-profile power	Frozen bounded result with explicit node uncertainty. Full-node AC and the same-system component-to-node bridge are not identified; do not collapse device/node/system/facility boundaries.

Authority / entity	What it is	What we use	Boundary / why it matters
NOAA / NCEI / MADIS	U.S. atmospheric observation systems	KS39 and KRDM temperature, dew point, pressure; derived wet-bulb KS39, KRDM, and tertiary KBDN temperature/dew-point/pressure with source-specific QC; derived RH/wet-bulb	Local external driver for cooling; station coverage differs by year.
City of Prineville utility public records (2026)	Customer utility-meter custodian	Facebook meter consumption, bulk/hydrant records, meter set/swap lifecycle, data-description note	Observed customer-service boundary. One unit = 100 ft ³ = 748 US gal. WATER-COMM/ADD'L WATER, bulk, SWR, WELL-METER-FOR-SEW, Trailer City, and Warehouse remain distinct until accounting relationships are identified.
OWRD	Oregon Water Resources Department	Water-use reports, water rights/PODs, well records, GWIS groundwater observations	Official Oregon water authority; municipal production, direct POD pumping, and campus withdrawal are distinct boundaries.
OHA Drinking Water Services	State drinking-water regulator/inventory	Prineville public-water-system source identities	Supports authoritative source/well cross-walks for the municipal system.
USGS	Federal water-science agency	HUC12 water use and Integrated Water Availability context	Regional modeled hydrologic/use context; not Meta-specific and not an aquifer node.
PacifiCorp / PACW	Regional utility / balancing-area context	PACW system identity and demand context	Regional power-system boundary; not campus electricity.
FERC	Federal electricity regulator	Form 714 PacifiCorp-West monthly demand/operations and East+West hourly shape	Historical regulatory evidence used to bridge pre-EIA hourly coverage.
EIA	Federal energy-statistics agency	EIA-930 PACW hourly data; EIA-860/923 generator/fuel/cooling data	Modern grid and generator context.
EPA	Federal environmental agency	eGRID regional emissions factors; CAMPD plant emissions	Regional accounting and plant emissions; not Meta-specific marginal impacts.
Oregon DEQ	State environmental regulator	Backup-generator permits, hours, fuel/emissions evidence	Onsite backup-generation context; permitted capacity is not IT load.
Crook County	Local permitting authority	Structural, electrical, mechanical, plumbing permits and inspections	Documents physical expansion and technology chronology; does not directly reveal IT MW or water use.

Appendix A lists the machine-generated Prineville source inventory; external validation streams are tabulated separately.

Three boundary rules that prevent false precision and double counting

1. **Regional/context quantities are not campus quantities.** For example,

$$E_{campus}^{Meta} \neq D^{PACW} \neq G_{plants}^{Oregon}.$$

2. **Different water boundaries are not summed unless a physical accounting relationship is known.** For example,

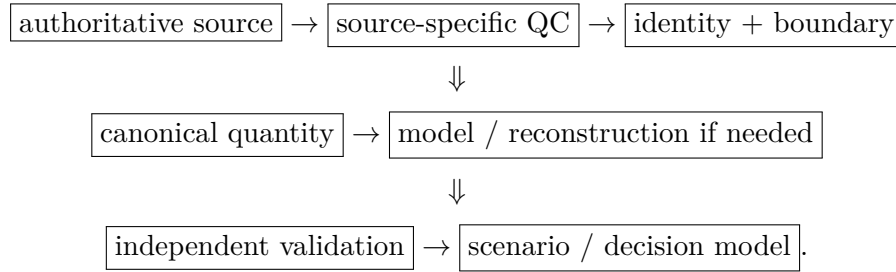
$$W_{campus}^{Meta} \neq W_{production}^{City} \neq Q^{direct\ POD} \neq W_{HUC12}^{USGS}.$$

The new customer records add another distinct boundary: $W_m^{City,service}$ is an observed municipal-service component, not a replacement for $W_m^{Meta,total}$. Bulk, SWR, WELL-METER-FOR-SEW, and direct-POD series are not summed into a campus total without a documented mass-balance/crosswalk.

3. **Geographic overlap does not imply physical connection.** A HUC12 is a geographic/surface-hydrology unit; it is not automatically a groundwater node, and proximity alone does not identify a serving well, generator, or aquifer connection.

3 The pipeline in plain language

The implementation is best understood as a sequence of evidence decisions rather than as a collection of scripts:



Step	What happens	Prineville example / rule
1. Gather	Prefer operator, regulator, or official measurement systems.	Meta disclosures, NOAA, OWRD, USGS, EIA/FERC/EPA, DEQ, Crook County.
2. QC	Check dates, units, duplicate records, missing-vs-zero, reporting conventions, and source-specific flags.	GWIS pumping/flowing observations are excluded from the state-model sample only when metadata support exclusion.
3. Crosswalk	Determine whether records refer to the same physical entity and boundary.	Airport wells share combined pumping; unresolved Vitesse PODs are not automatically mapped to nearby GWIS wells.
4. Canonicalize	Keep one defensible representation with provenance.	BLS and AMSL are paired representations of the same GWIS measurement, not two independent observations.
5. Model	Infer only quantities that are not directly observed.	Hourly IT load and subannual campus water require reconstruction/scenario assumptions.
6. Validate	Compare against independent data where possible; separate prediction from accounting closure.	FERC historical demand is checked against EIA-930; exact annual electricity closure is not called a forecast.
7. Scenario / decision	Use uncertain quantities only after their status is explicit.	Synthetic workload scenarios can explore interactions but cannot recover historical Meta telemetry.

The pipeline logic.

A multi-timescale problem

The model cannot honestly force every quantity onto the same time resolution:

Layer	Natural / available scale
Workload / server telemetry	seconds–minutes, but unavailable for Meta
Facility cooling / weather	minutes–hours
Regional grid	hourly
Meta campus disclosures	annual
City customer-service meter reports	monthly (2012–2026; 2026 partial)
OWRD pumping	monthly
Groundwater response	monthly / seasonal for the first reduced model
Siting / planning	annual / multi-year

The scientific challenge is therefore not merely filling missing cells; it is translating between time scales without pretending that annual measurements identify hourly internal states.

~~Prior weather wording used only KS39/KRDM; this is superseded.~~

Canonical model weather still covers the 2011–2024 local-calendar reconstruction window, but the current hierarchy is KS39 when QC-usable, KRDM as observational fallback/backbone, and bias-adjusted KBDN as a tertiary observational fallback for remaining required-driver hours. Only short bracketed primitive-variable gaps may be interpolated; RH and pressure-aware wet-bulb are recomputed after the final mixed-source selection.

4 From source to quantity to model

The provenance vocabulary used throughout the project is deliberately small:

Status	Meaning
REPORTED	Value officially reported by an operator or agency for the stated boundary.
MEASURED	Direct observational measurement such as weather or groundwater level.
DERIVED	Computed from observed/validated quantities using an accounting or physical relationship.
FITTED	Estimated from data using a documented statistical or physical model.
PROXY	Substitute for an unavailable target; useful structurally but not claimed as the target measurement.
SCENARIO	Explicitly assumed or simulated quantity used for sensitivity/counterfactual analysis.
EXTERNAL BENCH-MARK	External measured/validated evidence used to test model structure; not transferred numerically to the target site.
NOT IDENTIFIED	Scientifically relevant quantity/relationship that current evidence does not determine.
NOT NECESSARY	Not an active acquisition target for the present public-data pipeline; not necessarily scientifically unimportant.

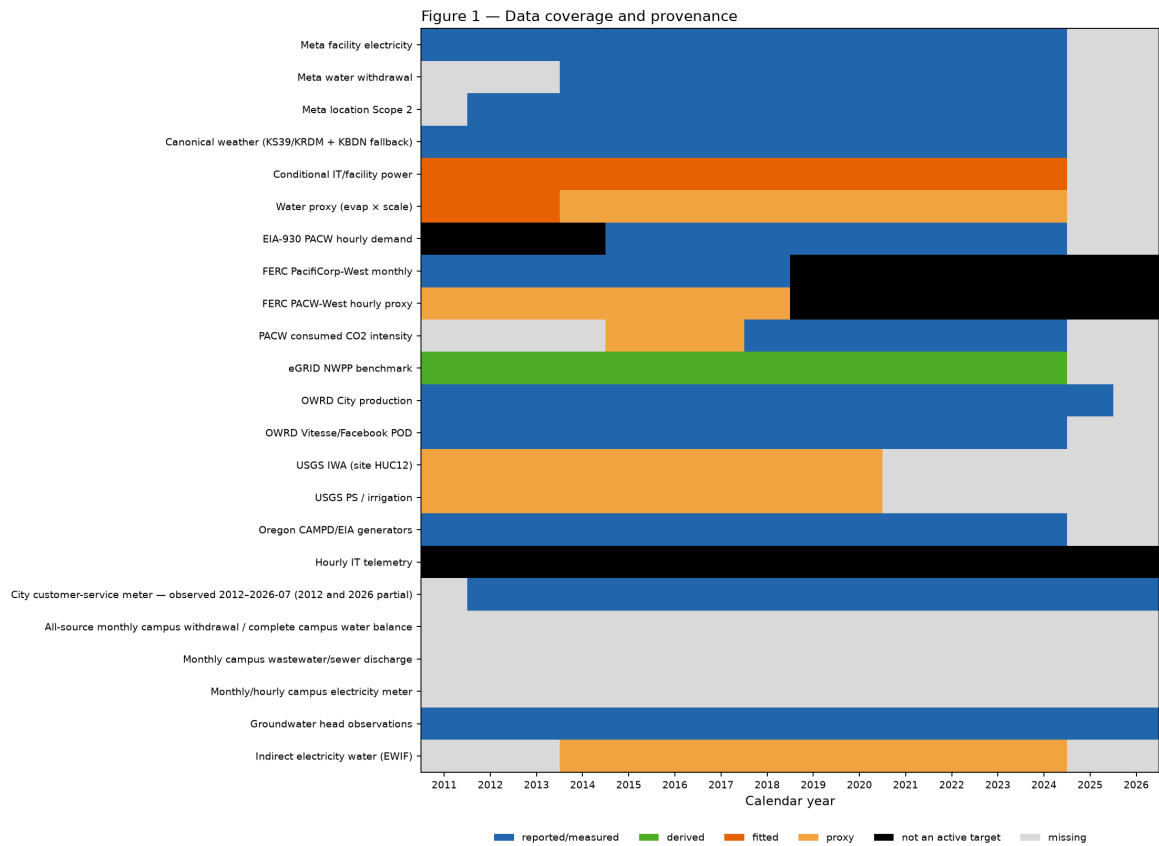


Figure 1: Data coverage and provenance, regenerated from the current Baseline-v2 reporting code. Evidence classes are reported/measured, derived, fitted, proxy, not-an-active-target, or missing over calendar years 2011–2026. Meta annual withdrawal is REPORTED; City WATER-COMM + ADD’L WATER is an OBSERVED customer-service component (2012, 2015, and 2026 partial); all-source monthly Meta/campus withdrawal remains NOT IDENTIFIED. Scope is coverage, not observation density or physical boundary equivalence.

Layer	Quantity	Source / authority	Status	Model role / current conclusion
Workload	x_t, u_t	none public for Meta	SCENARIO	Synthetic workload used only to create plausible operational shapes.
IT	P_t^{IT}	constrained by Meta annual facility MWh	FITTED/ SCENARIO	Latent internal state; not directly validated as telemetry.
Modern AI compute	$P_{w,N,t}^{compute}$	NLR/H100 + independent public evidence	frozen bounded external evidence	Workload/mode/scale conditioning is supported for compute components; full-node AC and facility-IT closure remain uncertain.
Facility	E_y^{fac}	Meta	REPORTED	High-confidence annual target.
Facility structure	$P_t^{IT}, P^{cool}, P^{fac}, PUE$	M100 ExaData [1]	EXTERNAL BENCHMARK	Chronological external validation: explicit weather dependence; PUE derived; cooling dominates M100 non-IT power; dynamic state form remains unidentified.
Facility archetype intensities	PUE_k, WUE_k, W^{cond}	Lei-Masanet [10]	external benchmark — / in-validation	First audit: IT-normalized climate technology intensity model with explicit conditioning-water components; annual envelope/RNG reproduction is the current gate.
Facility archetype structure	PUE_k, WUE_k, W^{cond}	Lei-Masanet [10]	EXTERNAL BENCHMARK; PARTIAL / closed	Structure and water-component semantics supported; quantitative adapter promotion blocked by the final reproducibility closure.
Facility overhead regime structure	$P^{overhead}, A^{arch}, S^{op}$	NLR/ESIF [4]	EXTERNAL BENCHMARK; PARTIAL / closed	Stationary IT+weather total-aux hypothesis fails; architecture/configuration and operating state must be allowed when spanning facility epochs.
Independent facility response	$p^{IT} \rightarrow p^{accessory}$	Frontier [3]	external benchmark — / QC-pending	First chronological result: affine IT-load model improves accessory-power MAE in all 9 folds; gap-safe coverage/energy correction is running.
Independent facility response	$P_t^{IT} \rightarrow p^{accessory}$	Frontier [3]	EXTERNAL BENCHMARK; closed	Corrected gap-safe audit leaves the all-9-fold load-dependence conclusion unchanged; no site-water/WUE inference.
Weather	T, T_d, T_{wb}, p	NOAA	MEASURED/ DERIVED	Strong cooling input layer.
Cooling	P_t^{cool}, Q_t^{cool}	gray-box	DERIVED	Mechanism/scenario quantity.
Campus water	W_y^{Meta}	Meta	REPORTED	High-confidence annual withdrawal target for 2014–2024; withdrawal is not consumption.
City customer-service water	$W_m^{City,service}$	City of Prineville utility records	REPORTED	Observed Facebook Data Center WATER-COMM + ADD'L WATER monthly service component; not total campus withdrawal and not a source allocation.
All-source monthly campus withdrawal	$W_m^{Meta,total}$	no closed source/accounting bridge	NOT IDENTIFIED	City service is one observed component; bulk/SWR/WELL/direct-POD/reuse/return/storage and scope relations do not identify the campus total.
Evaporation	W_t^{evap}	gray-box	DERIVED	Physical cooling mechanism, not total campus withdrawal.
Municipal/direct pumping	Q_t	OWRD	REPORTED	Water-system / groundwater forcing at explicit reporting boundaries.
Regional water context	HUC12 use / availability	USGS	PROXY	Regional context only; not a campus or aquifer measurement.
Grid demand	D_t^{PACW}	EIA + FERC	MEASURED/ PROXY	EIA later; FERC historical bridge earlier.
Carbon	CI_t	EIA / EPA eGRID	DERIVED/ PROXY	Regional average/accounting context.
Plant impacts	$G_{g,t}, CO_{2,g,t}, W_{g,t}$	EIA / EPA	REPORTED/ DERIVED	Generator-level context; no Meta-specific serving attribution.
Groundwater level	$BLS_{i,t}$ / reported AMSL	OWRD GWIS	MEASURED	QC-qualified individual-well observations; not a spatial/network head field.
Within-well relative head	$\tilde{h}_{i,t}$	derived from GWIS	DERIVED	$\tilde{h}_{i,t} = -(BLS_{i,t} - \overline{BLS}_i)$; physically signed target for the planned empirical benchmark.
Facility chronology	technology/events	Crook County	REPORTED	Documentary evidence of expansion/cooling regime changes.
Groundwater response / state	(β, A, B_R, B_Q)	not yet estimated	NOT IDENTIFIED	Empirical pumping response is the immediate benchmark; a reduced spatial/network state remains future work.
Decision variables	workload / siting / source choice	future	SCENARIO	Long-run optimization layer after response models are validated.

Master source–quantity–model map. Appendix B provides the complete machine-traceable version.

5 The model: general structure and current Prineville approximation

The literature suggests a coherent coupled spine. The Prineville implementation intentionally uses the simplest version supported by available public data.

5.1 Workload $\rightarrow$ IT power

A general operational model would begin with

$$E_t^{IT} = \mathcal{P}(x_t, u_t, \text{hardware}, \text{workload}) \Delta t.$$

Prineville: workload and server telemetry are not public. P_t^{IT} is therefore latent/scenario-based, not measured.

For the generic model, keep server/node power and the facility IT meter as distinct boundaries:

$$P_t^{IT, fac} = \mathcal{A}_h\left(\{P_{j,t}^{node}\}_{j=1}^{N_t}; \theta_h^{IT}\right).$$

M100 validates the need for this mapping: summed qualified node power strongly tracks the canonical facility IT meter but covers only roughly 74.5–80.9% of Tot_ict , so node power must not be equated mechanically with facility IT power.

~~The next compute-side experiment is already specified under `other_sources/it_power`: NLR, the 2026 H100/B200 dataset, and MLPerf Power provide complementary device/node/system boundaries. This is an experimental design only; no generic workload power map is yet validated.~~

The modern AI IT-power layer under `other_sources/it_power` now has machine disposition `FROZEN_BOUNDED_WITH_EXPLICIT_NODE_UNCERTAINTY`. Controlled CPU/H100 evidence supports workload-, mode-, scale-, and temporal-profile conditioning for compute-component power. It does not close full-node AC, other-node power, the production workload mix, or the node/system-to-facility-IT boundary.

5.2 IT power $\rightarrow$ facility electricity and cooling

~~The generic reading of the current scaffold is $P_t^{fac} = P_t^{IT} + P_t^{fan} + P_t^{other} + P_t^{cool, aux}$.~~

The externally validated generic accounting boundary should instead be written as

$$P_t^{fac} = P_t^{IT} + P_t^{cool} + P_t^{aux},$$

$$P_t^{cool} = f_k\left(P_t^{IT}, \mathbf{w}_t, s_t; \theta_k\right), \quad PUE_t = \frac{P_t^{fac}}{P_t^{IT}}.$$

Here k is a facility/cooling archetype, $\mathbf{w}_t$ is the relevant weather vector, and s_t is an optional latent/operating state. M100 strongly supports explicit weather dependence and derived PUE ; it does *not* identify universal coefficients, a universal weather variable, or a validated generic state equation.

~~For strategic/planning models, the minimum currently supported form is $P_t^{cool} = f_k(P_t^{IT}, \mathbf{w}_t; \theta_k)$.~~

~~The first Lei-Masanet audit supports a simpler *planning intensity backbone*:~~

Lei-Masanet supports the following *model class* as a climate-technology intensity abstraction, but its final reproducibility closure blocks promotion of the public implementation as a quantitative project adapter:

$$P_t^{fac} = P_t^{IT} PUE_k(\mathbf{w}_t, \theta_k), \quad P_t^{nonIT} = P_t^{IT} [PUE_k(\mathbf{w}_t, \theta_k) - 1].$$

This uses climate and cooling-technology archetype k to determine facility intensity; it does not claim that Lei-Masanet identifies nonlinear IT part-load behavior. The more general decomposition $P^{fac} =$

$P^{IT} + P^{cool} + P^{aux}$ remains the accounting architecture, and a dynamic $f_k(P^{IT}, \mathbf{w}, s)$ should be introduced only if later operational evidence makes it decision-relevant.

ESIF adds a separate caution on stationarity. When a model spans infrastructure/configuration epochs, the minimum supported predictive structure may require explicit architecture and operating-state variables:

$$P_t^{overhead} = f_k(P_t^{IT}, \mathbf{w}_t, A_t^{arch}, S_t^{op}; \theta_k), \quad P_t^{fac} = P_t^{IT} + P_t^{overhead}.$$

Here A_t^{arch} denotes facility/hardware/configuration epoch and S_t^{op} an operating/control state. This is a structural allowance, not an identified universal state equation and not permission to transfer ESIF coefficients.

With variable IT load, annual PUE must be energy-weighted:

$$PUE_y = \frac{\sum_{t \in y} P_t^{fac} \Delta t}{\sum_{t \in y} P_t^{IT} \Delta t},$$

not an unweighted arithmetic mean of hourly PUE. The two coincide in the upstream Lei–Masanet reproduction only because it normalizes hourly $P^{IT} = 1$. PUE accounting should follow ISO/IEC 30134-2 [5]; PUE remains a derived KPI, not an independently sampled primitive.

Prineville implementation. The annual IT scale is chosen so that

$$\sum_{t \in y} P_t^{fac} \Delta t = E_y^{Meta}.$$

Interpretation: this is an annual-closed reconstruction. The exact annual match is an accounting/calibration closure, *not* electricity forecast accuracy.

5.3 ~~Cooling → direct water~~ Cooling/facility → conditioning water → source water

~~The earlier generic water block mapped cooling directly to separate withdrawal and consumption response functions.~~

The first Lei–Masanet audit identifies an intermediate facility-side quantity more directly: modeled onsite *conditioning-water use*. For interval t ,

$$W_t^{cond} = E_t^{IT} WUE_k(\mathbf{w}_t, \theta_k),$$

where the upstream model’s water total includes cooling-tower evaporation, windage/drift, draw-off/blowdown, adiabatic/direct evaporative cooling, and humidification. This is an IT-normalized climate–technology intensity model; it does not separately identify source withdrawal, total consumption, return/discharge, or source shares.

Therefore the coupled project must retain an explicit second bridge:

$$W_t^{cond} + W_t^{other} \longrightarrow \{W_t^{with}, W_t^{cons}, W_t^{return}\} \longrightarrow \{W_t^{municipal}, W_t^{direct\ gw}, W_t^{reuse}, \dots\} \longrightarrow q_t^{gw}.$$

In particular, $WUE_k E^{IT} \neq$ groundwater pumping. M100 independently reinforces this boundary because its liquid-loop flow is circulation, not site water use.

With variable IT load, annual conditioning-water intensity must also be energy-weighted:

$$WUE_y = \frac{\sum_{t \in y} W_t^{cond}}{\sum_{t \in y} E_t^{IT}}.$$

The current project should keep this modeled conditioning-water intensity distinct from Meta withdrawal intensity and from any future source-resolved withdrawal/consumption accounting. ~~Facility-water gate~~

~~now running; annual Lei-Masanet envelope/RNG reproduction must pass before the adapter is used. The 2023–2024 Meta water holdout remains untouched. Source withdrawal/return/reuse still require separate evidence.~~

Lei-Masanet closure. Final disposition is PARTIAL and quantitative adapter promotion is NO/blocked. The climate-technology intensity structure and conditioning-water components remain useful evidence, but exact quantitative envelopes are not treated as a validated generic project adapter. The frozen Meta 2023–2024 water holdout was not read during the closure.

The current Prineville gray-box estimates a raw evaporative requirement from weather, airflow, heat load, and simplifying cooling assumptions. A parsimonious annual model then fits

$$\widehat{W}_y = s W_y^{evap}$$

or related energy/evaporation regressions to annual Meta withdrawal.

Interpretation: W^{evap} is a physical mechanism, not total campus withdrawal. The fitted scale absorbs unobserved non-cooling use, blowdown/reuse differences, architecture changes, and model error.

New Prineville observation. The City public-records integration now provides a directly observed monthly customer-service quantity

$$W_m^{City,service} = \sum_{r \in \mathcal{R}_{service}} W_{r,m}^{City}, \quad \mathcal{R}_{service} = \{\text{WATER-COMM, ADD'L WATER}\},$$

with source units of 100 ft³ per reported unit (748 US gal). This is a downstream municipal-service boundary. It is not $W_m^{Meta,total}$, not consumption, not municipal production, and not groundwater pumping. Bulk/hydrant, SWR METER, WELL METER FOR SEW, Trailer City, and Warehouse remain separate until their accounting relationships are identified.

5.4 Facility electricity → regional grid, carbon, and generator water

Attributionally,

$$\widehat{CO}_{2,t} = C I_t E_t^{grid}, \quad \widehat{W}_t^{ind} = E W I F_t E_t^{grid}.$$

Modern regional demand comes from EIA-930. Historical PACW-West hourly demand is reconstructed using FERC Form 714 monthly West totals plus East+West hourly shape, then validated against EIA-930 in overlap years.

Interpretation: these are regional accounting/context layers. They do not identify Meta’s exact serving generators or marginal grid response.

5.5 Water-source pumping → groundwater state

~~Earlier notation mapped groundwater forcing from a single $W^{dir,with}$ quantity.~~

The eventual coupled model should instead use the source-resolved groundwater share of facility withdrawal:

$$q_{n,\tau}^{dc} = \theta_{n,\tau}^{gw} W_{\tau}^{with},$$

where $\theta_{n,\tau}^{gw}$ is an explicitly identified groundwater/source allocation, not an assumed conversion from WUE or conditioning water. This forcing then enters a reduced groundwater state model such as

$$h_{\tau+1} = A h_{\tau} + B_R R_{\tau} - B_Q q_{\tau} + \varepsilon_{\tau}.$$

Prineville today: source shares and physical aquifer parameters are not identified. What we observe directly are well-specific GWIS BLS/AMSL measurements. For the first benchmark we derive a within-well,

physically signed relative head anomaly, $\tilde{h}_{i,t} = -(BLS_{i,t} - \overline{BLS}_i)$. This is enough to test a simple empirical pumping-response relationship, but it is not a common-datum spatial groundwater state and does not identify A, B_R, B_Q .

5.6 Simulation and future decisions

The stochastic workload module generates plausible workload/utilization shapes,

$$\text{synthetic arrivals} \rightarrow u_t \rightarrow P_t^{IT} \rightarrow \text{energy} / \text{cooling} / \text{water} / \text{carbon scenarios}.$$

Simulation is therefore an **experiment engine**: it can explore interactions under explicit assumptions, but it cannot recover missing historical telemetry or establish causal groundwater impacts.

Model-fidelity principle. The eventual planning model should compare a fixed-coefficient baseline, a validated climate-technology archetype model, and only if needed a higher-fidelity dynamic model by *decision relevance*—regret, changed site/cooling/source choices, capacity or groundwater violations, and environmental/equity outcomes—not only by predictive fit.

6 How the literature maps to the model

The project is deliberately not trying to implement every “state-of-the-art” model at once. The relevant question is which literature structure is justified by the data at each layer.

Literature family	What we inherit	Current Prineville approximation	What would justify the next upgrade?
Measured facility telemetry: M100 Exa-Data [1]	Measured IT/facility/cooling/weather relationships and accounting boundaries	External structural benchmark; no coefficient transfer to Prineville	Completed v3 closure; use findings as model constraints, then stop M100 fitting.
Thermal dynamics on M100 [2]	Same-data literature triangulation that dynamic cooling states can matter	Interpretive support only; not independent validation	Use independent facilities/models before adopting a dynamic state equation.
National/accounting: LBNL; Siddik et al. [7, 8] Facility PUE/WUE: Lei & Masanet [9, 10]	Explicit system boundaries, aggregate envelopes, spatial environmental accounting Climate- and technology-dependent coupled energy/water physics	Site-specific reported annual quantities + regional context Next generic archetype layer: cooling power, withdrawal, and consumption as climate/technology-dependent outputs First audit complete: IT-normalized PUE/WUE intensity backbone with explicit conditioning-water components	Use mainly as external order-of-magnitude / boundary checks. Annual 50-LHS×8760 reproduction; hidden-RNG gate; adapter tests; and Prineville 2022 weather translation are running; no Meta water holdout is read. Final closure PARTIAL; adapter promotion blocked. Retain structural/intensity evidence and water-component definitions, not a quantitative generic adapter.
Independent thermal/power benchmark: Frontier [3]	Measured compute/accessory power and coolant telemetry on a different liquid-cooled facility	Not yet integrated First chronological F0/F1/F2 test complete: IT-load term improves accessory-power MAE in all 9 folds	Gap safe expected time grid/energy QC correction is running; thermal field reproduction is not independent conservation validation. Corrected gap-safe audit is CLOSED; coverage is about 93.4% and the F1-vs-F0 qualitative conclusion is unchanged. Thermal reconstruction remains formula reproduction, not independent conservation.
Measured facility overhead: NLR/ESIF [4]	Measured IT, component overhead, PUE, and weather across a long facility history	Electrical overhead layer closes PARTIAL; stationary IT+weather total-aux hypothesis FAILS across the 2024 regime change	Use $f(P^{IT}, \mathbf{w}, A^{arch}, S^{op})$ as the minimum cross-epoch structural allowance; do not transfer coefficients or site PUE values.
Modern AI IT-power evidence: NLR / H100 B200 / MLPerf	Device, node, and system power at complementary boundaries	Source audit and smallest nested experiment are predeclared under other_sources/it_power	No generic workload $\rightarrow$ P^{IT} map is yet validated; preserve device/node/system/facility boundaries.
Modern AI IT-power evidence: NLR / H100 + independent public evidence	Workload-, mode-, scale-, and temporal-profile-conditioned compute-component power	Frozen bounded external layer with explicit node uncertainty	Use workload-specific compute scenarios; full-node AC, other-node power, production mix, and the facility-IT bridge remain unresolved.
Water footprint: Li et al.; Gupta et al.; Han et al. [11, 12, 13]	Direct vs indirect water; temporal water signals; peak public-water capacity	Annual direct water + partial regional generator-water diagnostics Annual Meta withdrawal + observed monthly City-service component + partial regional generator-water diagnostics	Monthly campus water, source shares, withdrawal/consumption/discharge, peak delivery. All-source monthly campus withdrawal, source shares, withdrawal/consumption/return, and peak delivery remain unresolved.
Operational scheduling: WACE; Google; WaterWise; TAPAS [14, 15, 16, 17] Siting/planning: PSCC project [20] Groundwater: MODFLOW / hydro-economic models [18, 19]	Workload conservation, spatial/temporal flexibility, carbon/water/thermal constraints Cost-carbon-water-equity planning backbone Pumping/recharge $\rightarrow$ head/storage dynamics; multi-sector allocation	Stochastic scenarios only; no actual workload optimization yet Long-run target, not current implementation First empirical well-level benchmark only	Real workload flexibility/capacity, validated environmental response signals, prices/costs. Credible source-specific water and groundwater response functions. Recharge/background forcing and sufficient hydrogeologic/response information for reduced-order or physical calibration.

Literature-to-model map: what is inherited now, what is simplified, and what evidence would justify more complexity.

7 External facility-model validation: M100 ExaData

M100 is used only as an EXTERNAL BENCHMARK for the missing internal facility layer. The canonical record is the v3 closure under `other_sources/m100/results/suitability_2021_v3_closure/`; Apr–Dec 2021

provide nine qualified full-facility months and eight expanding chronological test folds (May–Dec). The primary target is $P^{nonIT} = Tot - Tot_ict$; a validated cooling-power aggregate is a secondary target.

Claim tested	M100 experiment / evidence	Model consequence
Facility accounting boundary	Canonical Tot and Tot_ict pass strong internal checks; validated cooling components explain about 95.5–98.2% of M100 non-IT energy.	Use $P^{fac} = P^{IT} + P^{cool} + P^{aux}$; do not transfer the M100 cooling share.
Weather dependence	Adding weather to affine IT load improves held-out MAE in 8/8 expanding folds (about 42% mean improvement) and in 7/9 fixed within-month tests. Dry-bulb and wet-bulb variants both carry signal.	Weather must be explicit; do not hard-code a universal weather variable.
IT–weather interaction	The predeclared $P^{IT} \times T_{wb}$ interaction adds $\geq 5\%$ MAE improvement in 0/8 folds.	Do not require this interaction in the minimum generic model.
Observed free-cooling state	A regime/state enrichment is stably useful in only 2/8 folds and worsens most later folds.	Do not expose M100 <code>Free_Cooling_Status</code> as a required generic input.
Temporal state/dynamics	Static residuals have strong 1-h persistence; a one-step lag oracle helps strongly, but the tested recursive D1 simulator is worse than the static weather model in 7/8 folds.	Allow a future dynamic state for operational simulation, but its form is not identified by M100.
Node $\rightarrow$ facility IT boundary	Summed qualified node power tracks Tot_ict but covers only about 74.5–80.9% of that boundary.	Require a server/node-to-facility-IT aggregation map; do not equate GPU/node watts with facility IT watts.
Thermal transport	P^{IT} vs. $flow \times \Delta T$ is positively associated (Pearson roughly 0.45–0.89) on qualified periods; Sep–Oct twin-panel behavior is unresolved.	Qualitative thermal sanity only; no M100 thermal-kW closure or water-consumption claim.
Water use	No independently verified makeup/withdrawal/consumption meter was found; circulating flow is not site water use.	M100 cannot identify WUE, withdrawal, or consumptive-water parameters.

Bottom line. M100 falsifies a weather-independent fixed-efficiency facility model across changing conditions and supports an archetype-based response $P^{cool} = f_k(P^{IT}, \mathbf{w}, \text{optional state})$. It validates structure, not generic coefficients.

QC limits retained in the record. The processed tables support strong energy-integral/gap robustness but not a reconstructed $\geq 90\%$ raw source-sample coverage test. October’s reported-PUE channel is anomalous while canonical Tot/Tot_ict remains usable. The 2026 dynamic-model code was executable but did not numerically reproduce its bundled sample exactly, so it is treated only as same-data literature triangulation.

After the M100 reporting cleanup is merged, record the final git commit/hash here for immutable provenance.

8 ~~Facility energy–water archetype validation: Lei–Masanet + Frontier~~ External facility evidence beyond M100: Lei–Masanet, Frontier, ESIF, and transferability

~~The next external-validation layer asks whether the planning model can replace fixed PUE/water coefficients with climate technology intensities without pretending to recover a specific private facility. The first Lei–Masanet audit is complete; a bounded publication-scale annual reproduction is currently running. Frontier supplies a separate measured power/thermal check.~~

This layer is now mostly a *claim-restriction* exercise rather than an open-ended model search. M100 is frozen; Frontier’s corrected QC is closed; ESIF’s electrical overhead experiment is closed PARTIAL; Lei–Masanet’s final reproducibility closure is PARTIAL with adapter promotion blocked. Together they support a reduced structural family, but not a universal coefficient set.

PUE and WUE should be generated jointly from the same archetype, weather, and facility-parameter draw rather than sampled independently. LBNL is used only for qualitative technology/envelope triangulation because it shares modeling/authorship lineage with Lei–Masanet; it is not counted as independent validation.

Table 3: Final frozen external-validation synthesis. These streams constrain model structure at their stated boundaries; none supplies a calibrated Prineville facility/water model.

Evidence stream	Final status	Strongest supported result	Not identified / prohibited inference	Model consequence
M100	CLOSED FROZEN; tations	/ Measured accounting and weather structure; node power strongly bridges to, but is not equal to, facility IT.	No coefficient/PUE/cooling-fraction/water transfer; node/system is not facility IT.	Retain explicit IT + weather facility structure and boundary separation.
Frontier	CLOSED	Independent 9-fold evidence that IT load improves accessory-power prediction; corrected time-grid coverage 93.4%.	No WUE, withdrawal, site-water, or transferable thermal coefficient claim.	Retain IT-load-dependent accessory-power structure; thermal formula is an accounting reproduction.
Lei-Masanet	PARTIAL CLOSED; adapter blocked	/ Climate-by-technology intensity and onsite conditioning-water component structure are useful.	No quantitative adapter promotion, nonlinear IT-load twin, source/groundwater inference, or Meta calibration.	Use only bounded qualitative/archetype scenarios; preserve P_IT=1 intensity semantics.
NLR/ESIF facility overhead	PARTIAL CLOSED	/ Stationary IT+weather total-overhead mapping fails across the persistent 2024 HVAC/configuration regime shift.	No coefficient transfer or claim that one cross-epoch HVAC law is validated.	Include architecture/configuration and operating state; validate components and epochs.
Forest City v3	Qualitative PARTIAL; quantitative NOT VALIDATED	PAR-TIAL; Same-window climate/controller replay supports mechanism-specific transportability and station-robust zero summer DX.	Effective facility Delta-T, air-flow, cooling-only water, and numerical transfer remain unidentified; model not calibrated.	Transfer qualitative mechanisms only; engineering/utility records bind further progress.
Modern AI IT-power layer	FROZEN BOUNDED; node uncertainty	/ Controlled CPU/H100 compute-component energy is workload/mode/scale conditioned; independent H100 sanity check passes.	Full-node AC and same-system component-to-node bridge are unsupported; production workload mix and facility IT boundary remain incomplete.	Use bounded workload-specific compute scenarios plus explicit uncertainty for other-node, network, storage, and service power.

~~**Current annual validation gate.** The paper/code case crosswalk passed 12 focused tests. Six case×climate cells were then locked before inspecting reproduction errors and span wet/waterside, air-cooled, DX, economizer, hot/humid, and cold regimes. Each is evaluated using the paper-style 50 LHS facility samples ×8760 TMY hours. Hidden upstream RNG is tested separately against facility parameter uncertainty. Only if this gate passes do adapter tests and a Prineville-2022-weather translation proceed; the frozen Meta 2023–2024 water holdout is never read.~~

Closed Lei-Masanet verdict. The final reproducibility closure is PARTIAL and adapter promotion is NO/blocked. The physical/accounting/climate-technology structure remains useful, but exact quantitative envelopes are only partially compatible; no seed search, range tuning, weather substitution, or upstream-physics edit was used to rescue the result. The protected Meta 2023–2024 water holdout remained unread.

~~Replace this running-status paragraph with the canonical FOLLOWUP_V1_STATUS.json / FOLLOWUP_V1_SUMMARY.md result once the DAG completes. If the annual/RNG gate fails, keep Lei-Masanet as qualitative archetype evidence rather than a project adapter.~~

Forest City v3 transferability check. The frozen cross-site experiment ends with:

- QUANTITATIVE_PHYSICS_TRANSFER=NOT_VALIDATED;
- FACILITY_EFFECTIVE_DELTA_T=UNIDENTIFIED;
- COOLING_WATER_MAGNITUDE=UNIDENTIFIED;
- FRC1_COOLING_ONLY_WATER_MAGNITUDE=UNIDENTIFIED;
- MODEL_CALIBRATED=NO.

It provides strong same-window climate evidence and useful weather×controller decomposition, but only PARTIAL qualitative physics transfer. This reinforces the rule: transfer structure, not numerical coefficients.

9 What we have learned so far

1. The case is now empirically rich enough to ground the research question

Meta Prineville grew from a comparatively small early campus to a facility using electricity at the TWh/year scale. Reported campus electricity covers 2011–2024 (71,000 MWh to 1.73 TWh). Water withdrawal is reported from 2014. Location-based Scope 2 is reported from 2012; 2011 is not separately disclosed. Annual campus electricity, water withdrawal, and location-based Scope 2 evolve differently, which already shows that one single “size” proxy cannot explain every externality.

2. Annual electricity is much better identified than internal hourly operation

Reported Meta electricity provides strong annual scale, but public data do not reveal the true hourly workload, IT load, cooling mode, or facility component powers. The hourly electricity decomposition is therefore useful for physical/scenario reasoning, not historical telemetry recovery.

3. The current annual water model fails the most important predictive test

Improving the local weather layer did not solve annual water prediction. Training uses observations through 2022; 2023–2024 is a two-year chronological holdout ($N = 2$). The current conditional evaporation-scale model has holdout percentage errors of +128.9% (2023) and +100.6% (2024). A frozen training-mean baseline is closer (+8.6% and −40.4%). Energy-only and evaporation-only candidates also fail to beat those simple historical references.

With only two holdout years, this is a small statistical sample. The practical conclusion is nevertheless clear:

the current annual water model is a mechanism / falsification baseline, not a reliable predictor.

The failure is informative: energy plus simple evaporative weather physics is not sufficient to explain campus withdrawal. Missing drivers may include technology/mode, source/reuse strategy, blowdown, non-cooling use, or other operational changes, but the current data do not identify which one dominates.

3a. City utility meters convert part of the monthly water problem from modeled to observed

The v2 City integration adds 3,456 meter-month records covering 36 meters from 2012–2026, with 2026 treated as observed only through July. The canonical monthly response is the Facebook Data Center WATER-COMM + ADD’L WATER customer-service component. Bulk/hydrant, SWR METER, WELL METER FOR SEW, Trailer City, and Warehouse are retained separately and excluded from that response because their physical/accounting roles are unresolved.

~~Prior text mixed the native-support baseline MAE (8,035 m³) with common-support gray-box metrics; superseded below.~~

On the canonical identical common support ($n = 120$ months across ten complete years within 2014–2024; incomplete 2015 excluded), seasonal persistence is best by MAE (7,473.1 m³; RMSE 11,164.8 m³), while the frozen gray-box evaporation×scale candidate has MAE 11,728.1 m³ and RMSE 15,879.4 m³. The gray-box

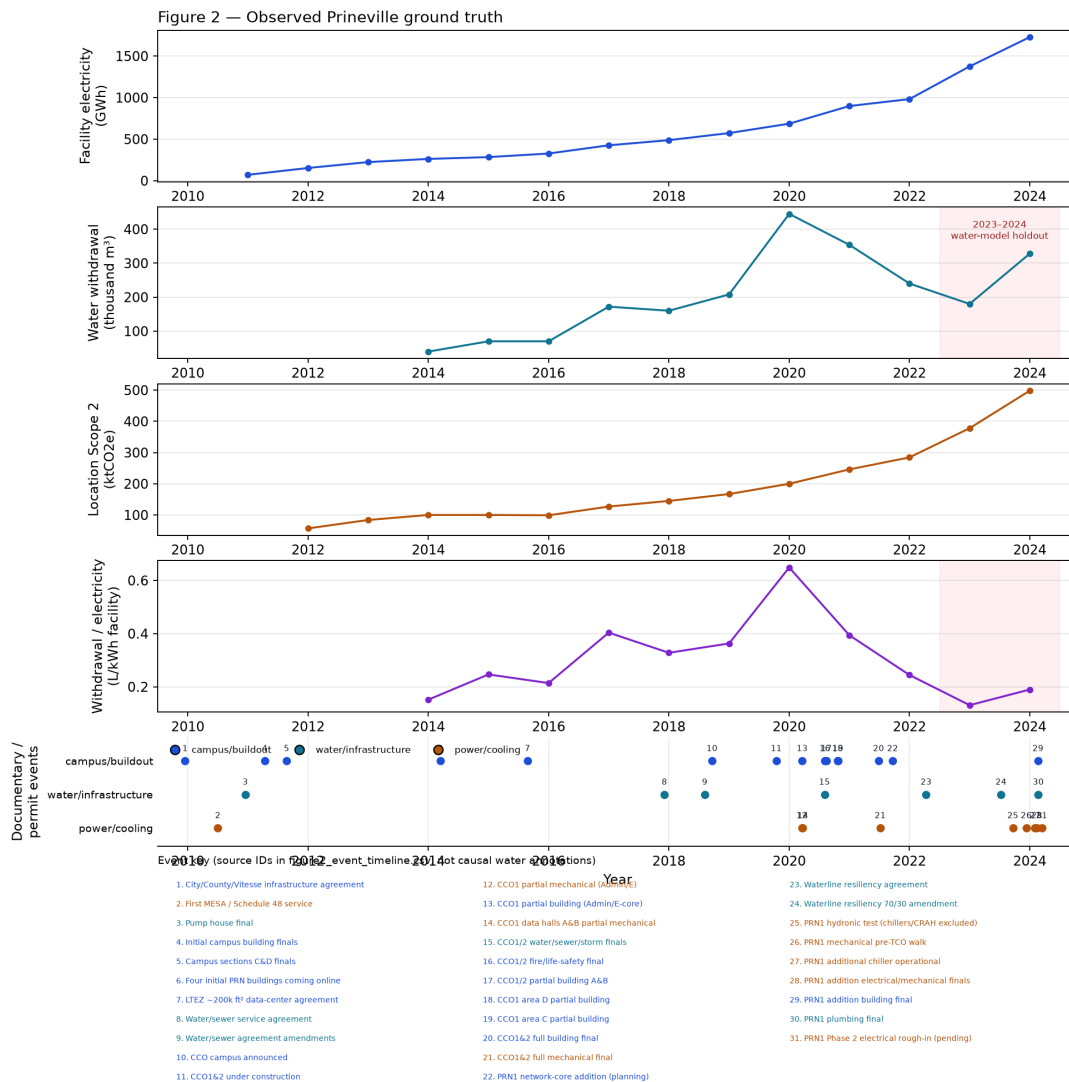
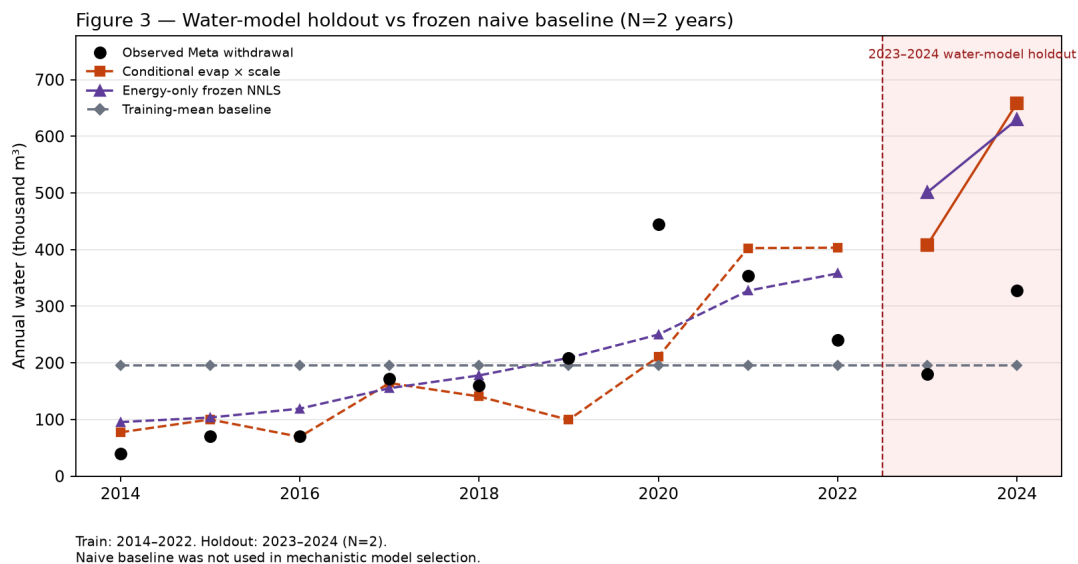


Figure 2: Observed annual Meta Prineville ground truth and intensity evolution. This is the empirical anchor for the case study. Evidence class: **REPORTED** operator disclosures; campus annual scope; electricity (n=14) (2011–2024, GWh), water withdrawal (n=11) (2014–2024, thousand m³), Scope 2 (n=13) (2012–2024, ktCO₂e), and derived withdrawal intensity (L/kWh facility). Documentary/permit events are chronology, not causal attribution.



Predictor	Holdout MAPE	2023 % error	2024 % error
Training mean	24.5%	+8.6%	-40.4%
Conditional evap x scale	113.8%	+126.7%	+100.8%
Energy-only NNLS	135.2%	+178.4%	+92.0%

Figure 3: Annual Meta water validation. The main message is comparative out-of-sample skill: current physics/energy predictors do not beat simple historical references on the two held-out years. **Evidence class:** frozen predictive validation; Meta annual campus-withdrawal boundary; holdout (n=2) (2023–2024); units thousand m³ and percent error. The figure is preserved unchanged and does not identify monthly water, consumption, source share, or mechanism.

retains seasonal-shape signal (pooled within-year-share correlation $r = 0.751$, $n = 132$ shape months) but overconcentrates use in summer: complete-year observed JJA shares span 0.365–0.544 versus 0.751–0.939 for the gray-box. This is an observed City-service validation, not an all-source campus-water validation.

Annual reconciliation is diagnostic only: City service is very close to Meta’s annual withdrawal in 2022, and City service + bulk is very close in 2023–2024, but that relationship is not stable in earlier years and is not promoted to a physical campus mass-balance identity. WELL METER FOR SEW versus OWRD direct POD has $n = 141$ overlapping reported months and Pearson $r = 0.0929$ ($n = 128$, $r = 0.1397$ when both are nonzero), so no source, master/submeter, or physical identity is asserted.

City-service monthly validation — observed component, not total Meta campus withdrawal

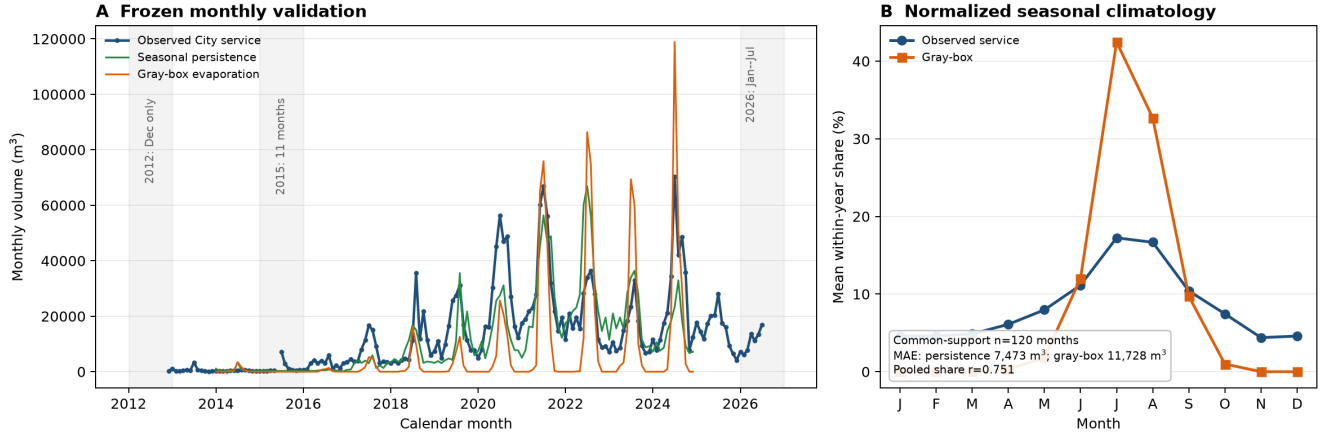


Figure 4: Frozen City-service monthly validation. Panel A compares the OBSERVED City WATER-COMM + ADD’L WATER customer-service component with the strongest frozen simple baseline and the frozen gray-box evaporation candidate; 2012 (December only), 2015 (11 months), and 2026 (January–July) are visibly partial and excluded from the $n = 120$ common-support ranking over ten complete years within 2014–2024. Panel B compares normalized mean within-year share (percent) for observed service and gray-box output. Units are m^3/month and percent share; scope is City service, not total Meta campus withdrawal.

Table 4: Frozen City-service validation metrics on identical common support ($n=120$ months across ten complete years in 2014–2024; incomplete 2015 excluded). Response is the observed City WATER-COMM + ADD’L WATER service component, not all-source Meta campus withdrawal.

Model	n	MAE (m3)	RMSE (m3)	SMAPE	Median AE (m3)	Information set
Month-of-year climatology	120	9,430.1	14,691.6	0.862	5,173.8	prospective_forecastable_baseline
Seasonal persistence	120	7,473.1	11,164.8	0.631	4,672.0	prospective_forecastable_baseline
Annual-electricity scale	120	8,692.5	12,973.2	0.648	6,236.8	contemporaneous_explanatory
Gray-box evaporation scale	120	11,728.1	15,879.4	1.557	8,460.5	contemporaneous_explanatory
Electricity + evaporation	120	8,518.4	12,686.2	0.612	4,809.5	contemporaneous_explanatory

The water boundaries used by this document are therefore kept explicit in the following main-body accounting table. “No” in the automatic-sum column is an identifiability restriction, not an assertion that the physical streams can never be related.

Table 5: City/Meta/OWRD water-boundary map. No row may be automatically summed with another without an identified accounting relationship.

Boundary / source	Canonical quantity	Status / coverage	Auto-sum?	Role / caveat
Meta annual withdrawal	Q_W_WITH W_Meta,y	REPORTED; 2014–2024 (11 annual values)	No	Annual campus disclosure; withdrawal, not consumption; no monthly allocation implied.
City WATER-COMM + ADD'L WATER service	Q_CITY_METER_SERVICE W_City,service,m	OBSERVED/REPORTED; No 2012-12 to 2026-07 (163 observed months; 2012, 2015, and 2026 partial)	No	Customer municipal-service component; model response; not all-source campus withdrawal.
City bulk/hydrant	Q_CITY_BULK_WATER	OBSERVED/REPORTED; No 2018-02 to 2026-08 (103 billing months)	No	Billing-month component with unresolved campus/use boundary; service+bulk is diagnostic only.
City SWR METER	Q_CITY_SWR_METER	OBSERVED/REPORTED; No identity unresolved; 2012-12 to 2026-07 (163 months)	No	Keep separate; naming/numerical proximity does not establish master/submeter or return semantics.
City WELL METER FOR SEW	Q_CITY_WELL_METER_SEW	OBSERVED/REPORTED; No identity unresolved; 2012-12 to 2026-07 (163 months)	No	Weak OWRD-POD correspondence; do not infer source, sewer, or master/submeter identity.
OWRD direct Vitesse/Facebook POD	Q_DIRECT_POD	REPORTED; 2010-10 to 2024-09 (450 reported POD-month rows)	No	Direct-POD reporting boundary; not total Meta withdrawal and not automatically additive to City service.
All-source Meta/campus withdrawal	W_Meta,total,m (coverage/report concept; no distinct registry row)	NOT IDENTIFIED; None	No	Service, bulk, SWR/WELL, direct POD, reuse/return/storage, lifecycle, and campus-scope relations do not close a mass balance.

4. Public regulatory data can reconstruct missing regional grid context

The FERC-constrained historical PACW proxy agrees well with EIA-930 during the mature 2016–2018 overlap ($r = 0.919$, $\text{NRMSE} = 0.083$). This makes pre-EIA regional demand usable as a *validated proxy*, not as directly observed West-hourly demand.

5. Independent permit evidence proves that facility technology is not temporally static

Crook County records document a PRN1 addition of approximately 82.7k ft² (reported range 82,273–82,736 ft²), IT-serving chilled-water/CRAH/chiller infrastructure, late-2023 commissioning/testing, and an additional chiller operational in early 2024.

This is important because a single cooling architecture over 2011–2024 is a simplification. It is **not** evidence that the permit event caused the 2023–2024 water-model failure, and it was not used to retune the holdout.

5a. Cross-site evidence supports structural, not quantitative, transfer

Forest City v3 is a deliberately conservative transportability experiment. Same-window observations show materially different climate, and the weather×controller decomposition confirms that cooling-regime occupancy depends jointly on climate and control logic. Yet transferred Lei–Masanet/ESIF calculations are not site calibration: qualitative physics transfer is PARTIAL, quantitative transfer is NOT_VALIDATED, cooling-water magnitude and effective facility ΔT remain UNIDENTIFIED, and MODEL_CALIBRATED=NO. The acquisition implication is also clear: engineering and utility records now outrank more weather data for improving site calibration.

6. Groundwater has crossed from “missing state” to “testable empirical layer”

The GWIS QC step processed 812 records, of which 800 contain numeric BLS measurements. After metadata-based exclusions, 796 are retained as potentially usable groundwater-state observations. Of these, 427 are explicitly STATIC and 369 are retained with ambiguous/unknown status.

Known problematic observations (for example PUMPING, FLOWING, INJECTING, DRY, or NOT MEASURED) are excluded only when metadata justify exclusion; surprising values are not deleted by magnitude alone.

The target is now physically signed:

$$\text{bls_anomaly}_{i,t} = \text{BLS}_{i,t} - \overline{\text{BLS}}_i, \quad \text{head_anomaly}_{i,t} = -\text{bls_anomaly}_{i,t}.$$

Three systems remain *estimation candidates* in the current Task-1 screen—meaning sufficient data to attempt a validated empirical response benchmark, not identified groundwater dynamics:

Heliport (SRC-GC), Millican (SRC-JA), Airport #2 (SRC-GB).

Airport pumping remains the accepted combined SRC-GA+SRC-GB group. The broad parameterized-aquifer model remains Class B / not ready. No groundwater-response model is fitted.

However, Heliport and especially Millican retain large REPORTED/UNKNOWN excursions that are not explained by available local method/status metadata. Measured groundwater observations are now available and several pumping/well systems contain enough overlap to attempt an empirical benchmark, but measurement-condition uncertainty remains material at some wells. The data are therefore *usable with explicit sensitivity analysis*, not “fully cleaned.”

10 What should we trust, and what remains unidentified?

Confidence ladder

Table 6: **Current confidence hierarchy. Confidence attaches to a quantity and boundary, not to a source name in isolation.**

Level		Components	Interpretation
High-confidence evidence	evi-	Meta annual ground truth; canonical NOAA weather; EIA modern grid; OWRD reported pumping; permit chronology; GWIS observations as observations	Use directly at their stated physical/accounting boundaries.
Observed City-service component		Facebook Data Center WATER-COMM + ADD’L WATER monthly utility-meter report; 2012–2026 with 2026 through July	Use as an observed customer-service boundary only. It does not close monthly campus withdrawal, source shares, sewer return, reuse, or groundwater forcing.
Externally—validated facility structure		M100: IT/facility/cooling accounting; weather-dependent cooling overhead; nonconstant derived PUE; persistent temporal structure	Use as structural constraints only. Do not transfer M100 coefficients, cooling fractions, control thresholds, or water quantities.

Level	Components	Interpretation
External facility-validation stack	M100 — structure; — Lei-Masanet climate/technology — intensity audit; — preliminary — Frontier IT-load/accessory-power validation	Use M100 as frozen structural evidence; treat Lei-Masanet as an intensity/archetype backbone only if the annual/RNG gate passes; treat Frontier result as preliminary until gap-safe QC closes. No generic coefficient transfer.
Closed/strong external structural evidence	M100 frozen structure; Frontier corrected load-dependence result	Use as model-form constraints at their boundaries; do not transfer site coefficients or water quantities.
Partial / transfer-restricted evidence	Lei-Masanet final PARTIAL closure; ESIF electrical PARTIAL closure; Forest City qualitative transfer PARTIAL	Use for archetype structure, regime/state allowance, and falsification. No quantitative generic adapter or cross-site coefficient transfer is validated.
Bounded compute-component evidence	Modern AI IT-power layer: frozen workload/mode/scale-conditioned CPU/H100 results	Use for bounded compute scenarios. Full-node AC, other-node power, production mix, and facility-IT aggregation remain explicit uncertainties.
Validated proxy / reconstruction	FERC historical PACW bridge; annual-closed facility-electricity profile	Useful because constraints/overlap validation are explicit; not raw telemetry.
Mechanism / scenario	cooling gray-box; stochastic workload; monthly water allocations; temporal carbon/water scenarios	Useful for sensitivity and counterfactuals, not historical identification.
Weak predictor	annual campus-water models	No demonstrated holdout advantage over naive references.
Weak monthly physical predictor	City-service gray-box level model	Seasonal persistence beats weather/evaporation models on MAE; gray-box shares are too summer-peaked. Use as mechanism/falsification evidence, not a promoted predictor.
Not yet identified	hourly Meta IT; monthly campus meters; — exact — source — shares; pumping-to-head causal response; aquifer dynamics; — exact — serving generators/marginal — impacts; costs/optimization	Requires new identification or data before strong claims/optimization.
Not yet identified	hourly Meta IT; all-source monthly campus withdrawal; source/return/reuse shares; generic quantitative PUE/WUE adapter; pumping-to-head causal response; aquifer dynamics; exact serving generators/marginal impacts; coupled planning/policy result	Requires source-accounting or response identification before strong coupled-planning claims.

Current project bottom line. Facility-model structure is sufficiently constrained for reduced planning representations, but generic numerical transfer is not validated. The observed monthly City-service component shifts the water problem from “invent a monthly shape” to source/boundary accounting. The sequential critical path is $W^{cond} \rightarrow \{W^{with}, W^{return}, \text{source share}\} \rightarrow q^{gw}$, then $q^{gw} + \{R, \text{background hydrology}\} \rightarrow h$, and finally deterministic M_0 -versus- M_1 decision replay.

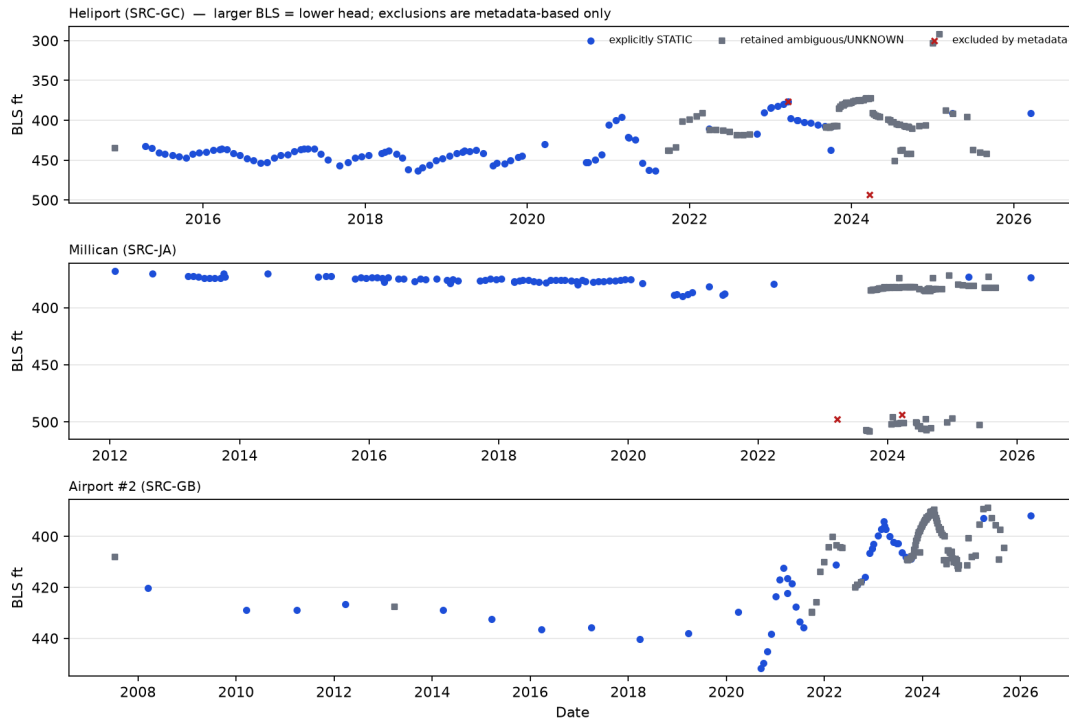


Figure 5: GWIS BLS at the three estimation-candidate wells. Blue circles are explicitly STATIC; grey squares are retained ambiguous/UNKNOWN; red crosses are metadata exclusions (not magnitude outliers). Larger BLS is lower hydraulic head. Evidence class: measured/reported GWIS observations; (n=488) plotted records across SRC-GC, SRC-JA, and SRC-GB, of which 483 are state-model eligible; units ft below land surface; individual-well scope. The preserved figure does not establish causal pumping response or a common-datum groundwater field.

Missing information by the scientific relationship it blocks

Relationship we want	Main missing / uncertain information	What it prevents today
Workload / hardware → facility IT	Modern AI workload power mapping and server/node-to-facility boundary parameters	Generic P^{IT} scenarios; NLR/H100/B200/MLPerf source audit and smallest experiment are designed but not yet validated.
Workload / hardware → facility IT	Full-node AC/other-node power, production workload mix, network/storage/service/idle load, and same-system aggregation evidence	The compute component is bounded, but a closed facility-IT scenario still requires explicit node/system/facility uncertainty.
IT + weather + technology → cooling power	Cross-technology archetype parameters and independent facility validation	Generic f_k calibration; next sources are Lei-Masanet/LBNL and Frontier.
IT + weather + technology → facility energy / conditioning water	Annual Lei-Masanet envelope reproducibility; hidden-RNG stability, and clean adapter semantics; Frontier gap-safe QC	Whether $PUE_k(\mathbf{w}, \theta_k)$ and $WUE_k(\mathbf{w}, \theta_k)$ can become the planning archetype backbone.
IT + weather + architecture/state → facility overhead	Validated quantitative parameterization that survives facility epochs; ESIF shows stationary IT+weather alone is insufficient	A universal numerical overhead law. For planning, retain bounded/archetype scenarios unless new independent evidence supports transferable coefficients.
Climate/technology → facility conditioning-water intensity	Quantitative generic adapter remains unavailable because Lei-Masanet promotion is blocked; cross-site transfer is not validated	Use the $W^{cond} = E^{IT}WUE_k$ structure as a model class, not as a validated universal parameterization.
Workload → IT / facility operation	workload telemetry, IT power, detailed cooling state/control	Historical hourly internal-state identification; this row is subsumed by the boundary-specific workload/hardware row above.
Cooling → campus water	monthly campus water, consumption vs withdrawal, discharge/return, cooling-loop operation, source/reuse shares	Reliable subannual water prediction and source-resolved accounting.
Conditioning water → source withdrawal / consumption / return	Campus source shares, municipal/direct-well delivery, reuse/reclaimed water, discharge/return, blowdown destination, storage/ASR accounting; City service is observed, but bulk/SWR/WELL/direct-POD/return/source relationships remain unresolved.	Groundwater forcing cannot yet be mapped.
W^{cond} $W^{with}/W^{return}/\text{source share} \rightarrow q^{gw}$	Service/bulk/SWR/WELL/direct-POD cross-walk; withdrawal-versus-consumption/return; reuse, blowdown, storage/ASR, meter lifecycle, and campus-scope accounting	An identified site-water mass balance and groundwater forcing series. City service alone does not close either quantity.
Pumping → groundwater	measurement-condition uncertainty; recharge/background hydrology; other pumping; physical T, S, S_y for full model	Causal/physical groundwater response and aquifer simulation.
$q^{gw} + \text{recharge/background} \rightarrow \text{groundwater response}$	Chronological response validation, measurement-condition uncertainty, recharge/background hydrology, other pumping; physical T, S, S_y for a later full model	A validated empirical head-response benchmark; no pumping-to-groundwater response is presently identified.
Facility load → grid externality	actual supply contract/serving attribution, marginal dispatch response	Meta-specific marginal carbon/water consequences.
Validated response functions → M_0 vs M_1 decisions	Closed source/GW forcing and response, workload flexibility, tariffs/costs, capacity constraints, and predeclared decision metrics	Credible replay of whether dynamic source/GW representation changes siting, technology/source choice, equity, or regret.

The important gaps are causal/identification gaps, not simply a list of missing files.

11 What simulation can and cannot do today

Simulation is useful when its role is explicit.

Useful now:

- explore plausible workload shapes and utilization patterns;
- ~~propagate IT load and weather through explicitly labeled facility/cooling archetypes, with PUE derived as an output;~~ propagate variable IT load through validated climate-technology intensity archetypes, with energy-weighted PUE/WUE and explicit conditioning-water components;

- compare explicit subannual water-allocation scenarios;
- study timing interactions with regional carbon signals;
- test future counterfactual technology/source/operating assumptions.

Not justified now:

- recovering Meta’s historical workload or hourly IT telemetry;
- ~~reconstructing actual monthly campus water without a meter/stronger model~~; recovering all-source monthly Meta campus withdrawal from the observed City-service component; the service boundary is observed, but the total/source/return balance remains unidentified;
- interpreting simulated pumping as the cause of an observed groundwater movement;
- treating scenario output as evidence of an actual historical cooling mode or source share;
- transferring M100 coefficients, free-cooling thresholds, cooling fractions, or circulating liquid flow as generic facility/water parameters;
- equating Lei–Masanet conditioning-water use/WUE with source withdrawal, consumption, or ground-water pumping.
- treating Forest City transferred Masanet/ESIF outputs as site calibration, or treating ESIF/M100 coefficients and regime thresholds as generic facility parameters;

12 Immediate next scientific decision

~~The next step was framed as one narrow groundwater-only empirical question.~~

~~**Updated sequencing after M100.** The immediate facility-model task is to validate climate technology response surfaces for cooling power, withdrawal, and consumption (Lei–Masanet/LBNL), with Frontier as an independent thermal benchmark; the modern workload/hardware block can proceed in parallel.~~

~~**Current sequencing.** The first Lei–Masanet audit is complete. The immediate facility gate is the predeclared annual PUE/WUE-envelope reproduction plus hidden-RNG test and Frontier gap-safe QC. If it passes, freeze a clean planning adapter $P^{fac} = P^{IT} PUE_k(\mathbf{w}, \theta_k)$, $W^{cond} = E^{IT} WUE_k(\mathbf{w}, \theta_k)$; if it fails, retain Lei–Masanet only as qualitative archetype evidence. In parallel, the modern workload/hardware $\rightarrow P^{IT}$ experiment is designed, and the pumping $\rightarrow$ groundwater benchmark remains a separate Thrust-A identification task.~~

~~**Current critical path.** The main bottleneck is no longer another facility-model search. M100 and Frontier are closed; ESIF’s electrical layer is closed PARTIAL; Lei–Masanet is closed PARTIAL with adapter promotion blocked; Forest City v3 does not validate quantitative transfer. The two immediate scientific tasks are therefore parallel: reconcile the observed City-service/bulk/SWR/WELL/direct-POD evidence and run the pumping $\rightarrow$ groundwater benchmark. Modern AI P^{IT} validation remains a supporting track.~~

Current critical path. The closed external-validation stack removes further generic facility-model search from the critical path. The work is now sequential: **(1)** reconcile site-water/source accounting from City service, bulk, SWR/WELL, direct POD, meter lifecycle, reuse/return/blowdown, and scope evidence without forcing closure; **(2)** translate the identified source shares into groundwater forcing q^{gw} ; **(3)** validate groundwater response using pumping plus recharge/background controls; and **(4)** replay the smallest deterministic source/GW-aware M_1 against static M_0 . The modern IT-power layer is already frozen bounded and remains a supporting uncertainty block, not the immediate bottleneck.

Groundwater-response stage (after forcing is defensible): does antecedent pumping add stable chronological predictive information about groundwater levels beyond seasonality, recharge/background hydrology, and trend?

To align monthly OWRD pumping with uneven GWIS measurement density, the first benchmark should use one target per well-month (for example, the median eligible head anomaly in that month).

Two samples should be pre-specified:

Primary: all metadata-eligible observations,
Sensitivity: explicitly STATIC observations only.

For SRC-GC, SRC-JA, and SRC-GB (with Airport pumping kept as the accepted combined SRC-GA+SRC-GB group), compare:

$$M_0 : \tilde{h}_{i,m} = \alpha_i + s_i(month_m) + \gamma_i m + \epsilon_{i,m},$$

against

$$M_1 : \tilde{h}_{i,m} = \alpha_i + s_i(month_m) + \gamma_i m + \beta_i X_{i,m}^{antecedent\ pumping} + \epsilon_{i,m}.$$

Use only completed prior-month pumping exposures; no same-month total that can contain pumping after the groundwater measurement. Validation should be chronological, not random.

Decision rule:

Stable incremental skill $\Rightarrow$ build a small reduced-order groundwater module.

No stable incremental skill $\Rightarrow$ add recharge/background controls before more complexity.

Only after the response models are credible should the project move toward coupled workload/source/siting optimization.

Ultimate fidelity test: compare the static PSCC-style M_0 against the smallest empirically defensible source-resolved/groundwater M_1 . Add higher-fidelity facility dynamics only if decision replay shows material regret, changed choices, or constraint violations.

13 Discussion questions for the meeting

The most useful advisor feedback is not on implementation detail, but on the modeling hierarchy:

1. Is the overall causal chain the right research abstraction: compute $\rightarrow$ facility $\rightarrow$ grid/water $\rightarrow$ environmental state $\rightarrow$ decision?
2. Is Prineville now sufficiently grounded to serve as the empirical validation case, despite missing internal telemetry?
3. ~~Should the next identification effort prioritize groundwater response, or is there a higher-value missing relationship (for example monthly campus water/source shares)? Is the next model-validation priority correctly split between generic facility energy/water archetypes and modern AI P^{IT} , and the parallel~~

pumping-to-groundwater-response benchmark? If the current Lei-Masanet annual/RNG-gate passes, is the IT-normalized climate-technology-intensity model the right planning fidelity, with higher-fidelity part-load/dynamic-cooling added only when it changes decisions? Given that quantitative Lei-Masanet promotion is blocked and Forest-City numerical transfer is not validated, should the first-coupled planner use bounded/archetype facility scenarios while prioritizing the now-observed City-service/source accounting bridge? Once the source-bridge and pumping-response benchmark are resolved, is that sufficient evidence to build the first deterministic M_1 for direct comparison against the static PSCC M_0 ?

4. Is the sequential critical path correct: site-water/source accounting $\rightarrow q^{gw} \rightarrow$ groundwater response $\rightarrow M_0$ -versus- M_1 replay?
5. What minimum source-accounting and groundwater-response validation gates are sufficient before the first deterministic M_1 decision replay?
6. Is the proposed progression—simple empirical response $\rightarrow$ reduced-order state model $\rightarrow$ later physical/optimization coupling—the right level of complexity?
7. Which eventual decision problem is most scientifically valuable: operations, source selection, siting, or multi-sector water allocation?

Document governance. This file is the advisor-facing glossary, traceability map, and evidence companion. The canonical project-wide scientific argument and research path remain in `main_documents/master.tex`; machine-readable/site-specific truth remains in the Prineville registries and experiment status artifacts. This separation is intentional so that the glossary does not become a second competing source of truth.

A Complete source traceability

The table below is generated from the current *Prineville* pipeline source catalog (4450 products). It is reference material, not meeting narration. External validation datasets such as M100 and Frontier are tracked separately from this site-specific source count.

Current public `main` identifies the Prineville package as Baseline v2. The generated v2 report lists 50 Prineville source products, 82 quantities, and 26 registered methods/models; the validated City-integration run reported 126 package tests PASS. The longtable below preserves the prior-generated catalog presentation.

At 08b0c112d8883eb4d23bb05d0d40e1a42c863593, deterministic reporting regenerates 50 Prineville source products, 82 quantities, and 26 registered methods/models byte-for-byte from the executable catalogs and frozen result claims. The rendered appendices below are generated from those current registries. External validation datasets remain outside the Prineville site-source count.

External model-validation sources are tracked separately from the site-specific count because their role is structural/archetype validation rather than Prineville measurement:

External source	Experiment / role	Current status
M100 ExaData [1]	Measured IT/facility/cooling/weather structure; node→facility boundary; temporal diagnostics	Closed/frozen structural benchmark.
Lei-Masanet 2022 + public code [10]	Climate technology PUE/WUE archetypes; explicit conditioning-water components; annual 50-LHS×8760 reproduction	First audit complete; bounded annual/RNG validation running.
Lei-Masanet 2022 + public code [10]	Climate technology PUE/WUE structure and conditioning-water components	Final reproducibility closure PARTIAL; quantitative adapter promotion NO/blocked; closed absent new upstream evidence.
Frontier [3]	Independent compute/accessory-power and liquid thermal context	First chronological load-dependence result positive; gap-safe QC correction running.
Frontier [3]	Independent compute/accessory-power and liquid thermal context	Corrected expected-time-grid audit CLOSED; load-dependence conclusion unchanged.
NLR/ESIF HPC Facility PUE Data [4]	Measured IT + facility-overhead components + weather; protected TEST	Electrical facility-overhead layer PARTIAL / closed; stationary IT+weather total-aux hypothesis FAIL.
Forest City v3 project experiment	Cross-site climate/controller robustness and transferred-model replay	Qualitative physics transfer PARTIAL; quantitative transfer NOT_VALIDATED; model calibrated = NO.
NLR / H100 B200 / MLPerf	Planned modern workload/device/node/system → P/T validation	Source audit + smallest experiment designed; no result yet.
NLR/H100 + independent public IT-power evidence	Workload/mode/scale-conditioned compute-component power and temporal profiles	Frozen bounded with explicit node uncertainty; full-node AC and facility-IT bridge incomplete.

Table 7: Machine-generated Prineville source inventory (50 products). Exact acquisition routes and local paths are preserved in the accompanying regenerated CSV. External validation datasets are not included in this site-specific count.

Source ID	Provider / product	Coverage / status	Role / boundary / caveat
META_2014_DISCLOSURE	Meta / Facebook: Facebook Sustainability Data 2014	2011-2014 vintage audit; annual; status=reported	historical source-vintage audit; first 2014 water disclosure Boundary: Prineville campus (site table). Caveat: Not the canonical later-year series; site totals are not hourly IT telemetry.
META_2015_DISCLOSURE	Meta / Facebook: 2015 Sustainability Data Disclosure	2011-2015 vintage; 2012 location Scope 2 used in canonical table; annual; status=reported	source-vintage audit; supplies 2012 location Scope 2 Boundary: Prineville campus (site table). Caveat: Annual corporate site table, not meters.
META_2016_DISCLOSURE	Meta / Facebook: 2016 Sustainability Data Disclosure	canonical early electricity (2011-2016); early water; Scope 2; annual; status=reported	canonical early electricity, water, location Scope 2, operational GHG Boundary: Prineville campus (site table). Caveat: Site annual totals. Do not treat as hourly IT workload.
META_2019_DISCLOSURE	Meta / Facebook: Facebook Sustainability Data 2019	canonical 2017-2019 electricity; 2015-2019 water; Scope 2; annual; status=reported	canonical mid-period electricity, water, location Scope 2 Boundary: Prineville campus (site table). Caveat: Site annual totals. Operational GHG definitions are vintage-sensitive.
META_2025_INDEX	Meta: Meta 2025 Environmental Data Index (FY2024)	2020-2024 electricity, water withdrawal, location Scope 2, operational GHG; annual; status=reported	canonical recent campus electricity, withdrawal, location Scope 2 Boundary: Prineville campus (site table). Caveat: Withdrawal is the disclosed site water quantity; consumption vs withdrawal is not separately identified for the campus.
META_2023_DATA_INDEX	Meta: Meta 2023 Environmental Data Index (methods)	WUE / withdrawal / consumption estimation definitions; methods through 2022; status=reported (methods, not the canonical numeric series)	definition/methods context for WUE and water accounting Boundary: company/site methodology. Caveat: Canonical numeric series come from 2016/2019/2025 vintages via build_targets.py, not this PDF.
META_ENGINEERING_2011	Meta Engineering: Designing a Very Efficient Data Center (2011-04-14)	2011 initial full-load PUE 1.07 and WUE 0.31 L/kWh; outside-air evaporative cooling; no chiller plant; design point (2011); status=reported design benchmark (not annual telemetry)	epoch-1 physics prior and PUE/WUE falsification point Boundary: Prineville initial design. Caveat: Design point, not measured annual PUE/WUE. Gray-box fan/other fractions are code priors, not values stated in this article.
META_EXPANSION_2021	Meta: Facebook's Prineville Data Center is Growing Again	2010 groundbreaking; 2021-03-18 expansion announcement (~900,000 sq ft; 11 buildings / ~4.6M sq ft upon completion); event; status=reported event	candidate capacity-state annotation only; completion dates not identified Boundary: campus. Caveat: Announcement is not a commissioning date or MW capacity series.
CITY_EOA_2026	City of Prineville: Economic Opportunities Analysis (2026 draft)	2010-2024; 11 data-center buildings constructed; planning/event; status=reported planning document	end-state campus-scale cross-check Boundary: city / campus. Caveat: Does not identify individual commissioning dates or MW.
CITY_WATER_SEWER	City of Prineville: City Water/Sewer system description	current; current description; status=reported system description	context for municipal ownership, wells/reservoirs, metering Boundary: municipal system. Caveat: Not a production time series. Facebook meter records are CITY_FB_* sources, not this page.
CITY_2024_CCR	City of Prineville: 2024 Consumer Confidence Report	2024; groundwater-only source statement; annual/system; status=reported	named well inventory / groundwater-only statement Boundary: PWS 00682. Caveat: Not monthly production. Not Meta campus delivery.

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Source ID	Provider / product	Coverage / status	Role / boundary / caveat
CITY_PUBLIC_RECORDS	City of Prineville: Public-records request mechanism	current request mechanism; ; status=not a dataset; request route only	request route; Facebook utility-meter package now ingested as CITY_FB_* sources Boundary: . Caveat: Meter-level Facebook water/sewer/bulk/lifecycle records were received in 2026. Municipal well production by well, ASR injection/recovery, and identified total campus wastewater return remain outstanding. City meters are not Meta annual withdrawal.
CITY_FB_METER_CONSUMPTION	City of Prineville: Facebook water/sewer meter consumption report	2012-12 through 2026-07 (2012 and 2026 partial); month (City: consumption month for ordinary meters); status=reported City utility-meter usage	primary observation of City-metered Facebook Data Center WATER-COMM + ADD'L WATER service Boundary: named Facebook accounts / meter ID at service location. Caveat: Customer-service boundary, not total Meta campus withdrawal. SWR METER and WELL METER FOR SEW remain unresolved identities; SWR physical_direction is unknown (label is not flow direction). 2026 Aug-Dec are not_observed_yet. CSV/TXT are mirrors; PDF is an evidence copy with sparse extractable text. Extra file FB Meters and Consumption.csv is a byte-identical local duplicate of (1).csv and is not a separate scientific source.
CITY_FB_BULK_WATER	City of Prineville: Facebook bulk/hydrant water billing records	2018-02 through 2026-08 bill dates; bill date (City: billed ~15th; reads ~10th); status=reported billing-convention usage	separate bulk component; excluded from city_metered_water_service_m3 Boundary: hydrant meter (no meter ID in the export). Caveat: Bill month is not claimed as physical consumption month. Duplicate bill dates are preserved. Not identified as ordinary cooling water.
CITY_FB_METER_EVENTS	City of Prineville: Facebook meter set/swap/pull history	City-compiled from ~2018 onward; incomplete earlier by City note; event date; status=reported lifecycle events (incomplete pre-2018)	lineage/anomaly audit; not consumption Boundary: meter ID. Caveat: Absence of a pre-2018 event is not proof the meter did not exist. Leading-1 ID crosswalk is inferred and not silently applied. Meter 1573376176 set 2026-02-15 appears in 2024 consumption; source not corrected.
CITY_UTILITY_DATA_DESCRIPTION	City of Prineville: City explanatory note on Facebook utility records	accompanies 2026 delivery; n/a; status=source metadata	authoritative City statements on units and timing Boundary: n/a. Caveat: Filename misspells Prineville. Metadata, not a time series.
OHA_PWS_00682	Oregon Health Authority / Drinking Water Services: Prineville City PWS 00682 inventory	official source IDs, well-log IDs, source status; current/historical status; status=reported inventory	municipal well/source identity for OWRD mapping and HUC12 join Boundary: source/facility inventory. Caveat: 13 wells unresolved (no official coordinates). Yancey Well #3 maps outside study HUC geography.
OWRD_WUR_QUERY	Oregon Water Resources Department: Water Use Report Query entity exports	City entity WY2010-2025; Vitesse/Facebook WY2011-2024 bundled; monthly where reported; status=reported water-use records (may be measured or estimated by reporter)	external municipal production and Meta/Facebook-associated direct POD evidence; not calibration targets Boundary: POD / Report-ID. Caveat: City production is not Meta delivery. Direct POD is not total Meta withdrawal. Missing months stay missing. DT14/DT18 unresolved.
OWRD_WUR_RULE	Oregon Water Resources Department: Water Use Reporting program rules	current; methods; status=methods documentation	reporting-requirement context Boundary: . Caveat: Not a numeric series.
OWRD_ASR	Oregon Water Resources Department: Aquifer Storage and Recovery program page	current; methods/system; status=methods documentation	ASR monitoring/accounting context only Boundary: . Caveat: No operational ASR injection/recovery series or groundwater-head model is implemented.
OWRD_ASR_2020_APP	Oregon Water Resources Department / City of Prineville: Prineville ASR Dedicated Well #1/#2 grant application (2020)	states 260 MG/y additional storage as application context; 2020 application / historical context; status=reported engineering/application document	groundwater-system intervention context Boundary: City ASR project. Caveat: Not a calibrated head/storage time series. Catalogued application PDF is still not present locally; 260 MG/y remains a secondary application citation. Local Crook County permit PDFs were scanned and do not contain this value.
OWRD_ASR_2020_ATTACH	Oregon Water Resources Department / City of Prineville: Prineville ASR 2020 attachments / 2018 feasibility study	hydrographs, aquifer properties, ASR pilot design as document context; numeric heads now come from GWIS, not digitized figures; 2018 study / 2020 attachment; status=reported engineering document	ASR engineering context; hydrographs not digitized because GWIS tabular levels exist Boundary: local aquifer / ASR pilot. Caveat: Catalogued attachments PDF is still not present locally. Local Crook County permit PDFs were scanned and yielded no T/S/Sy or pumping-test values. No groundwater dynamics model is implemented.
OWRD_GWIS	Oregon Water Resources Department: Groundwater Information System (GWIS) well, construction, lithology, and measured water-level exports	local export of eight unique wells after file-hash deduplication; numeric BLS/AMSL where reported; irregular measured water levels (datetime); status=measured well water levels and reported well-log construction/lithology	measured groundwater-level observations and official well identity/construction; not a groundwater dynamics model Boundary: individual wells (official GWIS coordinates where present). Caveat: Duplicate raw exports are provenance-only. Mixed NGVD1929/NAVD1988 datums are not converted. Unmatched wells remain candidate_unresolved. Combined OWRD pumping groups are not split.
NOAA_ISD	NOAA NCEI: Integrated Surface Database / Global Hourly product family	product family; Prineville uses KRDM files below; hourly/subhourly; status=measured (official station)	product documentation for the weather driver Boundary: station. Caveat: source_manifest still says station-ID verification needed; executable config uses KRDM 72692024230.
NOAA_GH_KRDM_FILES	NOAA NCEI: Global Hourly yearly station files: 72692024230 (KRDM / Roberts Field)	2011-2024 reconstruction window in processed weather_hourly.csv; hourly after processing; status=measured station weather; wet-bulb derived psychrometrically	exogenous weather driver for gray-box cooling/evaporation Boundary: KRDM station (Redmond Roberts Field; not on-campus). Caveat: Off-campus proxy for campus microclimate. source_manifest says files are not bundled; processed series is present.

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Source ID	Provider / product	Coverage / status	Role / boundary / caveat
NOAA_GH_KBDN_FILES	NOAA NCEI: Global Hourly yearly station files: 72063800224 (KBDN / Bend Municipal Airport)	tertiary gap-only fallback for hours where KS39 and KRDM cannot supply QC-usable T/Td in the 2011-2024 reconstruction window; hourly after processing; status=measured station weather; monthly additive transfer to KRDM climate; RH/Twb recomputed in canonical mix	tertiary gap-only weather fallback; not a replacement for KS39 or KRDM Boundary: KBDN station (Bend Municipal Airport; ~32 km from Prineville; not on-campus). Caveat: Off-campus. Used only when the KS39/KRDM hierarchy cannot supply valid primitives. Selection is independent of water/electricity model results.
NWS_KRDM	National Weather Service: Roberts Field / KRDM time series page	station identity confirmation; current; status=metadata	confirm nearby reference station KRDM Boundary: station metadata. Caveat: Not the numeric archive used by prepare_weather.py.
NOAA_MADIS_KS39	NOAA MADIS: KS39 / Prineville Airport METAR (MADIS)	QC-usable KS39 preferred in canonical weather from 2015-09-01 local; earlier KS39 is audit only; hourly after processing; status=measured METAR; station pressure derived from altimeter; RH/Twb recomputed in canonical merge	preferred near-site weather driver from 2015-09-01 local when MADIS QC-usable Boundary: KS39 station (Prineville Airport; not on-campus). Caveat: Not campus microclimate. MADIS QC is not relaxed. 2016 pressure fallback is genuine NOAA QC, not a parser bug.
USGS_NWAA_API	U.S. Geological Survey: National Water Availability Assessment Data Companion API	model-specific (see product rows); monthly; status=official modeled	acquisition route for HUC12 water-use and IWA products Boundary: HUC12. Caveat: Not Meta meters. No CONUS dump. Do not use after documented product end dates.
USGS_NWAA_IWA	U.S. Geological Survey: IWA assessment outputs CONUS 2025	2009-10 through 2020-09; monthly; status=official modeled	regional IWA streamflow, cumulative consumption, availability identity, SUI Boundary: HUC12 (site 170703051002; local 9; same-HUC8 52). Caveat: availab = strflow - consum is an internal identity; not independent hydrologic validation. consum is cumulative upstream+local. Cannot support 2021-2024.
USGS_NWAA_PS_CU	U.S. Geological Survey: Public-supply consumptive-use model (pscutot)	2009-01 through 2020-12; monthly; status=official modeled	modeled public-supply CU context; not Meta-specific Boundary: HUC12. Caveat: Do not add pscutot into IWA consum. Missing after 2020-12.
USGS_NWAA_PS_WD	U.S. Geological Survey: Public-supply withdrawals model (pswdtot/pswdgw/pswdsw)	2000-01 through 2020-12; monthly; status=official modeled	modeled public-supply withdrawal context Boundary: HUC12 / HUC8 plus extra hydro-near HUC12. Caveat: Not City OWRD production and not Meta withdrawal. Missing after 2020-12.
USGS_NWAA_IRR_WD	U.S. Geological Survey: Crop irrigation withdrawals model (irrwtdot)	2000-01 through 2020-12; monthly; status=official modeled	modeled irrigation withdrawal context Boundary: HUC12. Caveat: Competing-sector context only. Missing after 2020-12.
USGS_NWAA_IRR_CU	U.S. Geological Survey: Crop irrigation consumptive-use model (irrcutot)	2000-01 through 2020-12; monthly; status=official modeled	modeled irrigation consumption context Boundary: HUC12. Caveat: Do not add irrcutot into IWA consum. Missing after 2020-12.
USGS_WBD	U.S. Geological Survey: Watershed Boundary Dataset HUC12 (The National Map)	study HUC geography; current WBD service vintage used in the download; status=official geospatial	study HUC geography and municipal well point-in-polygon Boundary: HUC12 polygon. Caveat: Site HUC12 170703051002 is site_point_huc12; campus polygon is unverified. Direct Facebook PODs have no coordinates.
EIA930	U.S. Energy Information Administration: EIA-930 / Real-Time Operating Grid	bundled workbook 2015-07-01 through reconstruction window 2024-12-31; hourly; status=reported BA operations; some EIA imputations flagged	regional demand/generation/interchange context; hourly consumed CO2 intensity where present Boundary: balancing authority PACW. Caveat: PACW is not campus electricity. Consumed CO2 intensity begins 2018-07. Not a Meta-specific marginal-emissions model.
EIA_PACW	U.S. Energy Information Administration: PacifiCorp West balancing-authority dashboard	2015-07-01 onward in bundled workbook; hourly; status=reported	PACW identity for the EIA-930 extract Boundary: BA PACW. Caveat: Same workbook as EIA930; dashboard is the acquisition route, not a second numeric source.
FERC_FORM_714	Federal Energy Regulatory Commission: Form 714 Annual Electric Balancing Authority Area and Planning Area Report	2011-2018 annual filings present under data/raw/ferc_form_714/; monthly West NEL/ops; hourly East+West Schedule 2 demand; status=reported FERC filings; West hourly reconstruction is a FERC-constrained proxy	2011-2018 reported PacifiCorp-West monthly demand/operations; East+West hourly shape; candidate pre-EIA PACW hourly proxy; EIA-930 overlap validation Boundary: PacifiCorp-West BA (monthly); PacifiCorp East+West combined planning area (hourly). Caveat: East+West hourly is not PACW-West. FERC does not provide consumed CO2 intensity, fuel mix, hourly interchange, campus electricity, or marginal emissions. Does not replace data/processed/pacw_hourly.csv.
EPA_EGRID	U.S. Environmental Protection Agency: eGRID detailed US-customary workbooks	eGRID2010-2023 workbooks covering model years 2011-2024; annual vintage (lagged map for odd years / 2024); status=official modeled annual generation-weighted output rates	methodology/accounting consistency benchmark = Meta MWh x NWPP output rate versus Meta location Scope 2; not fully independent external validation Boundary: eGRID subregion NWPP (ZIP 97754). Caveat: Not hourly campus intensity, not PACW demand, not market/REC accounting. 2024 uses eGRID2023.
EPA_EGRID_POWER_PROFILER	U.S. Environmental Protection Agency: Power Profiler ZIP Code Tool v14.2	ZIP 97754 uniquely NWPP; Subregion 2/3 blank; eGRID2023 geography; status=official lookup	campus consumption-location subregion assignment Boundary: ZIP 97754 → NWPP. Caveat: Geography lookup only; rates come from eGRID vintages.
OREGON_BER_RENEWABLE	Oregon Department of Energy: Biennial Energy Report, Renewable Energy chapter	2018 PacifiCorp Schedule 272 unbundled REC agreement; policy/event 2018; status=reported policy/event	market-vs-location carbon accounting breakpoint annotation Boundary: PacifiCorp / Facebook contract context. Caveat: Does not change location-based eGRID/PACW physical accounting.
EPA_CAMPD_OR	U.S. EPA CAMPD: Oregon hourly emissions extract (CEMS)	2011-2024 in processed campd_or_unit_hourly.csv (QC: years 2011-2024 present); hourly; status=measured CEMS (posted mass); gross generation derived as Gross Load x Operating Time	Oregon generator emissions/generation; no site-to-generator attribution Boundary: Oregon CAMD Facility ID x Unit ID. Caveat: Does not identify which generators served Meta Prineville. Posted CO2/SO2/NOx mass are not multiplied by Operating Time.

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Source ID	Provider / product	Coverage / status	Role / boundary / caveat
EPA_EIA_CROSSWALK	U.S. EPA / EIA: EPA/EIA unit crosswalk	used to join Oregon CAMPD units; unmatched units retained; crosswalk vintage as bundled; status=official crosswalk (exact / modified_fuzzy / official_manual methods in QC)	join CAMD units to EIA identifiers without exploding emissions Boundary: CAMD unit ↔ EIA plant/generator/boiler. Caveat: Cardinality includes one_to_many and many_to_many; emissions are not copied onto generator IDs.
EIA_860	U.S. Energy Information Administration: EIA Form 860 annual generator inventory	2011-2024 model years; annual; status=reported inventory	Oregon generator characteristics / operating-retirement dates Boundary: Oregon plants/generators after filter. Caveat: National archives filtered to Oregon after read. No campus attribution.
EIA_923	U.S. Energy Information Administration: EIA Form 923 generation and fuel	2011-2024; monthly where respondent frequency=M; annual-only respondents flagged; status=reported (monthly reporters); EIA-allocated months for annual respondents are not treated as observations	Oregon net generation/fuel; CAMPD vs EIA-923 reconciliation diagnostic Boundary: Oregon plants after filter. Caveat: Annual respondents' published monthly Netgen is not a monthly observation. No campus attribution.
EIA_COOLING	U.S. Energy Information Administration: EIA standardized cooling-detail / Schedule 8 cooling operations	standardized cooling water used for 2014-2024; 2013 Schedule 8 volumes where reported in million gallons; 2011-2012 flow rates not converted; annual/monthly as published; status=reported where present	Oregon thermoelectric cooling water; generator-side water intensity where generation is associated Boundary: Oregon cooling systems after filter. Caveat: 2011-2012 native units not converted. Negative official consumption is not clipped and is not used for intensity. Not Meta indirect water.
ODEQ_AIR_07_0037	Oregon Department of Environmental Quality: Vitesse LLC (Meta/Facebook Prineville) air permit 07-0037-ST-01	text-extractable ARs include 2012, 2014-2017, 2020, 2022-2023; scan-only 2019/2021 ARs not OCR'd; AR_2013/AR_2018 missing per remaining_actions; monthly hours/emissions/fuel where extractable from annual reports; status=reported permit/annual-report quantities; source tests stored separately	onsite backup generation/emissions independent of grid Scope 2 Boundary: onsite backup generators at the Prineville campus. Caveat: Scan-only PDFs unused. Proposed units are not treated as active. Nameplate MW is not IT capacity. PSEL/PTE is not actual emissions.
ODEQ_GHG	Oregon Department of Environmental Quality: Oregon GHG electricity-delivery workbooks	Pacific Power sheet processed; other GHG workbooks treated as provenance unless they contain Vitesse 07-0037 observations (none identified); annual as published in the workbook; status=reported utility-delivery GHG	utility-territory GHG context; not Meta location Scope 2 Boundary: Pacific Power Oregon deliveries (not campus). Caveat: Not campus electricity and not onsite backup. Kept separate from eGRID/PACW.
CROOK_COUNTY_PERMITS	Crook County ePermitting / inspection summaries: Vitesse/Meta Prineville building permit inspection-summary PDFs	2010 onward inspection summaries in the curated table; event / opened and final-or-CO dates; status=reported permit/inspection chronology	capacity-epoch and commissioning-support chronology; PRN1 2023/2024 transition is interpretation/scenario evidence, not holdout tuning; no inferred exact sqft or MW Boundary: campus parcels / building_id. Caveat: Opened date is an ePermitting proxy, not necessarily permit issuance. PRN1 addition area is an unresolved ~82.7k-ft ² range (82,273/82,723/82,736); exact_final_area missing. Amp/circuit counts are not converted to MW. Pipe diameter is not consumption. 2023-2024 permit transition is not used to retune gray-box or water holdout.
DOCUMENTARY_EVIDENCE_BUNDLE_V1	City of Prineville / Oregon PUC / Vitesse LLC: Prineville documentary/regulatory evidence bundle v1 (21 PDFs; curated seed CSVs)	21 unique documents; 48 curated evidence rows; event / legal chronology 2009-2026; status=reported documentary/regulatory facts; not meters	identity/boundary, interconnection/service-capacity, transmission topology, and water-infrastructure/legal context; not calibration Boundary: PRN campus, CCO campus, City service, regional transmission. Caveat: PRN vs CCO Meta reporting boundary is unresolved. MW quantities are not campus load. Pipe diameter/cost/funding is not campus water. Permit v2 package is already integrated separately and is not rescanned here.

B Complete quantity / model traceability

The current registry contains 7682 glossary quantities and 1826 registered methods/models. Implemented quantities are listed first; quantities that remain NOT IDENTIFIED are grouped compactly; methods are classified by role so that accounting identities, reconstructions, fitted predictors, simulations, and QA screens are not all read as predictive models.

~~The purple rows below synchronize the advisor-facing glossary with the validated v2 registry and the latest closed external experiments. The machine-generated v2 registries remain authoritative if a row count or implementation detail later diverges from this manually maintained companion.~~

The quantity and method/model appendices below are rendered directly from the regenerated current registries. External-validation experiments remain in the separate synthesis table and do not inflate the Prineville registry counts. All-source monthly campus withdrawal is explicit in the coverage and boundary

ledgers but is not a distinct row in the current 82-row quantity registry; the registry's Q_W_WITH is annual.

Table 8: Machine-generated model quantity registry (82 quantities).

Quantity	Unit / resolution	Provenance / status / boundary	Definition / source / limitation
Q_ARRIVALS: Workload arrivals	Unit: scenario work counts (not kWh); time: hourly in stochastic ensemble	scenario; implemented as generative scenario only; boundary=SCENARIO_WORKLOAD	New work arriving in interval t . No public scheduler traces exist; the stochastic module draws scale-free Cox-process counts. Source/model: none (no telemetry); src/stochastic_conditional_simulation.py. Limitation: Hourly IT workload traces are not public.
Q_EXECUTED: Executed workload	Unit: scenario work units; time: hourly (stochastic)	simulated; implemented as generative scenario only; boundary=SCENARIO_WORKLOAD	Work executed in t from a stable aggregate queue. Stochastic only; conditional reconstruction has no queue. Source/model: src/stochastic_conditional_simulation.py. Limitation: Not recovered from public data.
Q_CAPACITY: Compute capacity	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Usable compute capacity. Not identified from public sources. Source/model: . Limitation: No hardware inventory, nameplate IT MW time series, or scheduler availability.
Q_UTILIZATION: Utilization	Unit: index (scale-free); time: hourly (stochastic)	scenario; scenario index only; boundary=SCENARIO_WORKLOAD	Share of usable capacity that is active. Stochastic module uses a scale-free utilization_index from executed work; it is not measured $u=x/K$. Source/model: src/stochastic_conditional_simulation.py. Limitation: No rack/server utilization telemetry.
Q_BACKLOG: Backlog	Unit: scenario work units; time: hourly (stochastic)	simulated; implemented as generative scenario only; boundary=SCENARIO_WORKLOAD	Unfinished work in the stochastic aggregate queue. Source/model: src/stochastic_conditional_simulation.py. Limitation: Not a scheduler queue.
Q_DELAY_SLA: Delay and SLA	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Job delay and class delay limits. Not implemented. Source/model: . Limitation: No job timestamps or SLA constraints in the pipeline.
Q_P_IT: IT power	Unit: MW; time: hourly (reconstructed/simulated); annual scale	fitted; implemented as fitted latent (conditional) / simulated-then-closed (stochastic); boundary=META_LATENT_IT	Electricity used by IT equipment. Conditional reconstruction uses one latent constant-within-year IT-power scale so gray-box facility energy closes to reported annual facility MWh. Stochastic uses a workload shape then the same annual closure. Source/model: Meta annual facility electricity + NOAA weather + gray-box. Limitation: Hourly IT load is not public. Within-year variation is assumed (conditional) or scenario (stochastic).
Q_E_IT: IT energy	Unit: MWh; time: hourly sum $\rightarrow$ annual	derived; implemented as derived from fitted P_IT; boundary=META_LATENT_IT	Hourly IT power summed to MWh. Derived from fitted/simulated P_IT. Source/model: fitted P_IT. Limitation: No measured IT energy series.
Q_WEATHER: Weather state	Unit: degC; percent; Pa; time: hourly	measured; implemented; boundary=META_CAMPUS	Canonical KS39/KRDM hourly weather for local calendar years 2011–2024. QC-usable KS39 METAR is preferred from 2015-09-01 local; KRDM is the backbone and gap-fill; KBDN NCEI Global Hourly is a tertiary gap-only fallback. RH and wet-bulb are recomputed after the final T/Td/pressure combination. Source/model: NOAA_GH_KRDM_FILES / NOAA_MADIS_KS39 / NOAA_GH_KBDN_FILES. Limitation: Off-campus station bias unquantified.
Q_IT_HEAT: IT heat	Unit: MW_th (implicit); time: hourly	derived; implemented as physics identity inside gray-box; boundary=META_LATENT_IT	Operational-scale IT heat. Gray-box uses $Q_{IT} \approx P_{IT}$ to size airflow: $m_{air} = P_{IT} \times 1e6 / (c_p \times \Delta T_{server})$. Source/model: fitted P_IT. Limitation: No loop heat measurements.
Q_COOL_LOAD: Cooling load	Unit: MW_th (implicit); time: hourly	derived; partial: $Q_{cool} \approx Q_{IT}$ baseline; boundary=META_LATENT_IT	Heat that must be removed. Code uses the IT-heat airflow requirement; building heat and dU/dt are not modeled. Source/model: gray-box. Limitation: No measured cooling load or building heat.
Q_THERMAL_STATE: Thermal state vector	Unit: degC; time: hourly	derived; partial (t_{supply_C} only); boundary=META_LATENT_IT	Rack/inlet/GPU temperatures. Only supply-air temperature from the evaporative controller is computed. Source/model: gray-box + weather. Limitation: No rack/inlet telemetry. return_air_C prior is declared but not used in simulate().
Q_COOLING_MODE: Cooling technology/mode	Unit: categorical; time: hourly	derived; implemented as rule-based mode; boundary=META_LATENT_IT	Gray-box discrete mode: outside_air_or_winter_mix / partial_evap / full_evap from weather vs 25 C target. Source/model: gray-box + canonical KS39/KRDM weather. Limitation: No BMS mode telemetry. PRN1 2021–2024 permits document chilled-water/CRAH/chiller infrastructure and a late-2023 $\rightarrow$ early-2024 PRN1 transition; the gray-box remains a simplified common-architecture baseline and is not returned from that evidence because 2023–2024 are the water-holdout years.
Q_COOLING_CONTROL: Cooling control	Unit: degC / effectiveness; time: hourly implicit	scenario; partial (fixed priors); boundary=META_LATENT_IT	Setpoints/fan speeds/flows. Only the 25 C supply target and evaporative effectiveness prior are used. Source/model: code priors in Params. Limitation: Exact cooling setpoint/control telemetry is not public.
Q_P_COOL: Cooling power	Unit: MW; time: hourly	derived; implemented as fan + evaporative-aux fractions of IT power; boundary=META_LATENT_IT	Gray-box has no chiller COP. Cooling-related electricity is $p_{fan} + p_{evap_aux}$ only. Source/model: gray-box priors. Limitation: No component meters. Constant COP is not used (and would be the wrong technology).
Q_P_FAC: Facility power	Unit: MW; time: hourly reconstructed; annual reported	derived; implemented; boundary=META_CAMPUS	$P_{fac} = P_{IT} + P_{fan} + P_{other} + P_{evap_aux}$. Hourly series is reconstructed; annual integral is forced to Meta electricity. Source/model: gray-box + annual electricity closure. Limitation: No hourly feeder power.
Q_E_FAC: Facility electricity (annual reported)	Unit: MWh; time: annual	reported; implemented; boundary=META_CAMPUS	Meta-disclosed Prineville annual facility electricity. Hard observation for calibration. Source/model: META_2016_DISCLOSURE / META_2019_DISCLOSURE / META_2025_INDEX via build_targets.py. Limitation: No monthly/hourly campus electricity observations.

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Quantity	Unit / resolution	Provenance / status / boundary	Definition / source / limitation
Q_PUE: PUE	Unit: unitless; time: hourly and annual	derived; implemented as derived KPI; boundary=META_LATENT_IT	P_fac/P_IT from the gray-box. Annual modeled PUE ≈ 1.060 . 2011 design full-load PUE 1.07 is a design/assumption consistency (falsification) check, not independent validation. Source/model: gray-box. Limitation: No simultaneous IT and facility metering.
Q_HEAT_REJ: Ambient heat rejection / reuse	Unit: unavailable; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Heat reuse is not modeled. Coarse balance $Q_{rej} \approx P_{fac}$ is not stored as an output. Source/model: . Limitation: No recovered-heat meters; reuse not assumed.
Q_WUE_DIR: Direct WUE	Unit: L/kWh (facility withdrawal intensity); design WUE L/kWh_IT; time: annual	derived; withdrawal/facility-kWh derived; ISO WUE unavailable; boundary=META_CAMPUS	ISO/IEC 30134-9 WUE = consumption / IT energy. The pipeline does not observe site consumption. It reports withdrawal / facility electricity as a derived intensity and 2011 design WUE 0.31 L/kWh as a benchmark. Source/model: Meta annual water and electricity. Limitation: No site consumption (needs discharge). Denominator is facility electricity, not IT energy.
Q_A2_WUE: a2 fixed-approach / fixed-cold-water WUE curves	Unit: L/kWh heat removed; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Gupta et al. generic cooling-tower WUE vs wet-bulb. Not implemented in executable code. Source/model: . Limitation: Code uses air-side evaporative humidification physics instead of a2 tower curves.
Q_W_EVAP: Raw evaporative water	Unit: m ³ /h (hourly); m ³ (annual sum); time: hourly	derived; implemented; boundary=META_LATENT_IT	Gray-box humidification water from humidity-ratio increase at constant moist-air enthalpy, converted m ³ /h. Source/model: gray-box + weather + fitted P_IT. Limitation: Not equal to withdrawal, consumption, blowdown, or municipal delivery.
Q_MAKEUP_BLOWDOWN: Makeup / blowdown / drift	Unit: unavailable; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Cooling-tower mass-balance streams. Not implemented. Source/model: . Limitation: No cycles of concentration, blowdown, or drift data.
Q_COC: Cycles of concentration	Unit: unavailable; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Not identified. Source/model: . Limitation: No water-chemistry or blowdown data.
Q_W_CONS: Direct water consumption	Unit: m ³ ; time: unavailable at site	unavailable; boundary=NOT_IDENTIFIED	Water not returned to the local system. Not identified: no discharge/sewer series and Meta tables used here are withdrawal. Source/model: . Limitation: Site consumption is not publicly reported as a separate campus series. City SWR METER is not identified as total return flow.
Q_W_WITH: Direct water withdrawal (Meta annual)	Unit: m ³ ; time: annual (2014-2024)	reported; implemented; boundary=META_CAMPUS	Meta-disclosed Prineville annual water withdrawal. Hard observation. Not monthly. Source/model: META_2016_DISCLOSURE / META_2019_DISCLOSURE / META_2025_INDEX. Limitation: Monthly Meta campus-total withdrawal is still not observed. City utility meters observe a customer-service component, not this annual campus total. 2011-2013 missing.
Q_CITY_METER_SERVICE: City-metered Facebook Data Center utility/service water	Unit: m ³ ; time: month (consumption month per City note)	reported; implemented; boundary=CITY_UTILITY_CUSTOMER_DELIVERY	Sum of City Facebook Data Center meters whose rate code begins WATER - COMM or ADD'L WATER, in observed months. City-metered utility/service water, not total Meta withdrawal. Source/model: CITY_FB_METER_CONSUMPTION. Limitation: Does not include bulk, sewer, WELL METER FOR SEW, Trailer City, or Warehouse. Not groundwater.
Q_CITY_BULK_WATER: City Facebook bulk/hydrant water (billing month)	Unit: m ³ ; time: bill date / bill month	reported; implemented; boundary=CITY_UTILITY_CUSTOMER_DELIVERY	Hydrant-meter bulk water organized on the City billing convention. Bill month is not claimed as consumption month. Source/model: CITY_FB_BULK_WATER. Limitation: Billing convention $\neq$ consumption month. Not part of city_metered_water_service_m3.
Q_CITY_SWR_METER: City SWR METER volume (unresolved identity)	Unit: m ³ ; time: month	reported; implemented as reported volume; identity unresolved; boundary=CITY_UTILITY_CUSTOMER_DELIVERY	Volume on meters with rate code SWR METER. Raw label is preserved. Flow direction is unknown. Not wastewater return, total discharge, or a consumptive-use offset. Source/model: CITY_FB_METER_CONSUMPTION. Limitation: Sewer volume is not automatically total return flow.
Q_CITY_WELL_METER_SEW: City WELL METER FOR SEW volume (unresolved identity)	Unit: m ³ ; time: month	reported; implemented as reported volume; identity unresolved; boundary=CITY_UTILITY_CUSTOMER_DELIVERY	Volume on meters with rate code WELL METER FOR SEW. Not identified as sewer discharge or as groundwater withdrawal. Source/model: CITY_FB_METER_CONSUMPTION. Limitation: Rate-code label is not quantity identity.
Q_CITY_TRAILER_WATER: Facebook Trailer City water-coded usage	Unit: m ³ ; time: month	reported; implemented as reported volume; campus role unresolved; boundary=CITY_UTILITY_CUSTOMER_DELIVERY	Water-coded meters on the Facebook Trailer City entity. Not identified as main-campus operations. Source/model: CITY_FB_METER_CONSUMPTION. Limitation: Excluded from city_metered_water_service_m3.
Q_CITY_WAREHOUSE_WATER: Facebook Warehouse water-coded usage	Unit: m ³ ; time: month	reported; implemented as reported volume; campus role unresolved; boundary=CITY_UTILITY_CUSTOMER_DELIVERY	Water-coded meters on Facebook Warehouse at 542 SW Commerce Ct. Not identified as main-campus operations. Source/model: CITY_FB_METER_CONSUMPTION. Limitation: Excluded from city_metered_water_service_m3.
Q_WATER_PROXY: Hourly/annual water-withdrawal proxy (conditional)	Unit: m ³ ; time: hourly proxy; annual prediction	proxy; implemented; boundary=META_CAMPUS	Train-only global multiplicative scale on raw evaporation. Selected model is global (not one-break). Holdout years use the frozen train scale. Source/model: raw evaporation $\times$ fitted scale. Limitation: Proxy is not a meter. Poor holdout means the frozen scale does not travel. Current fitted/holdout numbers are applied at report generation from canonical outputs.
Q_CHI: Consumptive fraction	Unit: unavailable; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	W_cons / W_with. Not identified. Source/model: . Limitation: Needs consumption and withdrawal on the same boundary.
Q_THETA: Source-share vector	Unit: unavailable; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Share of withdrawal by municipal / surface / groundwater / reclaimed. Not identified for Meta campus water. Source/model: . Limitation: No campus source-share meter. Direct POD is not total withdrawal.
Q_REUSE_WATER: Onsite water reuse	Unit: unavailable; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Not identified. Source/model: . Limitation: No reuse meters.
Q_ADD_MDD: Average-day and maximum-day demand	Unit: unavailable as daily; time: unavailable	unavailable; boundary=NOT_IDENTIFIED	Paper-c planning metrics. Not computed as campus daily demand; only annual withdrawal exists. Source/model: . Limitation: No daily campus withdrawal.

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Quantity	Unit / resolution	Provenance / status / boundary	Definition / source / limitation
Q_CITY_PROD: Municipal / City production	Unit: m ³ ; time: monthly	reported; implemented; boundary=PRINEVILLE_PWS	OWRD accepted City of Prineville municipal source-group production. External water-system evidence. Source/model: OWRD_WUR_QUERY. Limitation: Do not compute prediction error vs Meta withdrawal.
Q_DIRECT_POD: Vitesse/Facebook direct POD use	Unit: m ³ ; time: monthly where reported	reported; implemented; boundary=VITESSE_DIRECT_POD	OWRD reported use at Meta/Facebook-associated direct groundwater PODs. Source/model: OWRD_WUR_QUERY. Limitation: No official POD coordinates. Incomplete months stay missing.
Q_IRRIGATION: USGS irrigation water	Unit: m ³ /month (converted from mgd); time: monthly	proxy; implemented as regional context; boundary=HUC12_LOCAL_USE	Modeled HUC12 crop irrigation withdrawal and consumptive use. Source/model: USGS_NWAA_IRR_WD / USGS_NWAA_IRR_CU. Limitation: Ends 2020-12. Not campus water.
Q_IWA_STRFLOW: IWA cumulative streamflow	Unit: mm/month and m ³ /month; time: monthly	proxy; implemented; boundary=HUC12_ROUTED_HYDROLOGY	USGS IWA cumulative streamflow (upstream+local) in mm and m ³ per month. Source/model: USGS_NWAA_IWA. Limitation: No 2021-2024 IWA.
Q_IWA_CONSUM: IWA cumulative consumption	Unit: mm/month and m ³ /month; time: monthly	proxy; implemented; boundary=HUC12_ROUTED_HYDROLOGY	USGS IWA cumulative consumption (upstream+local, not local-only). Source/model: USGS_NWAA_IWA. Limitation: Do not treat as Meta consumption.
Q_IWA_AVAIL: IWA surface-water availability	Unit: mm/month and m ³ /month; time: monthly	derived; implemented; boundary=HUC12_ROUTED_HYDROLOGY	USGS IWA availability field. Code documents availab = strflow - consum as an internal identity. Source/model: USGS_NWAA_IWA. Limitation: Never call this independent hydrologic validation.
Q_IWA_SUI: IWA supply-use index	Unit: fraction (native); time: monthly	proxy; implemented; boundary=HUC12_ROUTED_HYDROLOGY	USGS IWA SUI field for the site HUC12. Source/model: USGS_NWAA_IWA. Limitation: Ends 2020-09. Not head or storage.
Q_USGS_PS: USGS public-supply withdrawal/CU	Unit: m ³ /month; time: monthly through 2020-12	proxy; implemented as regional context; boundary=HUC12_LOCAL_USE	Modeled HUC12 public-supply water use. Not City OWRD and not Meta. Source/model: USGS_NWAA_PS_WD / USGS_NWAA_PS_CU. Limitation: Missing after 2020-12.
Q_GRID_DEMAND: PACW grid demand / interchange / generation	Unit: MWh; time: hourly	reported; implemented; boundary=PACW_BA	Balancing-authority operations from the bundled PACW workbook. Source/model: EIA930. Limitation: Starts 2015-07. Demand reported has gaps; adjusted series fills some EIA imputations.
Q_FERC_WEST_MONTHLY: FERC PacifiCorp-West monthly NEL and operations	Unit: MWh; MW; time: monthly 2011-2018	reported; implemented; boundary=PACW_BA	Reported FERC Form 714 PacifiCorp-West monthly net energy for load, net generation, net interchange, monthly peak, and monthly minimum. Regional BA evidence, not campus electricity. Source/model: FERC_FORM_714. Limitation: Ends 2018. Not hourly reported PACW demand.
Q_FERC_EW_HOURLY: FERC PacifiCorp East+West combined hourly demand	Unit: MW; time: hourly 2011-2018, hour-ending local Pacific prevailing time	reported; implemented; boundary=PACIFICORP_EAST_WEST_PLANNING_AREA	Reported FERC Form 714 Schedule 2 hourly demand for the PacifiCorp East+West combined planning area. Used only as intramonth shape. Never labeled PACW-West. Source/model: FERC_FORM_714. Limitation: Not PACW-West. Time convention is FERC hour-ending local Pacific, not EIA-930 hour-ending UTC.
Q_FERC_PACW_HOURLY_PROXY: FERC-constrained PACW-West hourly demand proxy	Unit: MW; time: hourly 2011-2018	proxy; implemented as diagnostic/candidate pre-EIA proxy; boundary=PACW_BA	FERC-only affine reconstruction: within each month, East+West hourly shape scaled to West monthly NEL with nonnegative b_m chosen against West peak and minimum. Exact monthly energy closure to NEL_W. Not reported PACW hourly demand and not campus electricity. Source/model: FERC_FORM_714. Limitation: Does not supply consumed carbon, fuel mix, hourly interchange, or campus electricity. Does not overwrite pacw_hourly.csv.
Q_RENEWABLE_CF: Renewable capacity factor	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Site or BA renewable CF for scheduling. Not implemented as a campus model input. Source/model: . Limitation: No campus renewable production model.
Q_GEN_OR: Oregon generator generation and emissions	Unit: MWh; tonnes; time: hourly CAMPD; monthly EIA-923	measured; implemented as regional generator tables; boundary=OREGON_GENERATOR	CAMPD gross generation and posted CO2/NOx/SO2; EIA-923 net generation/fuel. Oregon-only pilot. Source/model: EPA_CAMPD_OR / EIA_923. Limitation: Generator-to-Meta attribution is not identified.
Q_GEN_WATER: Generator cooling water	Unit: m ³ (from million gallons); time: as published (2014-2024 standardized product)	reported; implemented as Oregon tables; boundary=OREGON_GENERATOR	EIA cooling-product water for Oregon plants where reported. Source/model: EIA_COOLING. Limitation: Not coupled to campus electricity as generator attribution. Partial-coverage regional EWIF is a separate derived table.
Q_EWIF: Average electricity water intensity factor	Unit: m ³ /MWh; time: annual aggregation of monthly plant observations	derived; implemented as partial-coverage regional diagnostic; boundary=OREGON_GENERATOR	Generation-weighted cooling-water intensity over Oregon EIA-923 plant-months that are cooling-intensity-eligible with positive cooling-associated generation and non-missing water. Partial-coverage regional diagnostic, not a complete grid EWIF and not Meta generator attribution. Source/model: EIA_COOLING. Limitation: Complete Oregon-grid EWIF is not identified because plants without cooling water are not assumed zero. Generator-to-Meta attribution is not identified.
Q_MWF: Marginal water factor	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	$\partial W_{\text{power}} / \partial E_{\text{load}}$. Not implemented (no dispatch model). Source/model: . Limitation: No dispatch/commitment model.
Q_W_IND: Indirect electricity water	Unit: m ³ ; time: annual	proxy; implemented as regional-average proxy only; boundary=OREGON_GENERATOR	Regional-average proxy = partial-coverage Oregon cooling EWIF $\times$ Meta campus MWh. Not generator-attributed campus water and not an identified Meta-serving mix. Source/model: EIA_COOLING $\times$ Meta electricity. Limitation: Generator-resolved attribution is not identified. This proxy must not be labeled Meta-specific indirect water.
Q_W_OPER: Total operational water footprint	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Direct consumption + indirect. Unavailable because both components are unidentified at the required boundaries. Source/model: . Limitation: Cannot form identified W_oper: campus consumption is not identified, and indirect water is only a regional-average proxy rather than generator attribution.

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Quantity	Unit / resolution	Provenance / status / boundary	Definition / source / limitation
Q_SCOPE2_META: Meta reported location-based Scope 2	Unit: tCO2e; time: annual	reported; implemented; boundary=META_CAMPUS	Disclosed Prineville location-based Scope 2 (tCO2e) where present. Source/model: Meta disclosures. Limitation: 2011 Meta-reported location Scope 2 is not separately disclosed. An eGRID location-based accounting proxy exists and must not be labeled Meta-reported.
Q_SCOPE2_EGRID: eGRID NWPP physical carbon benchmark	Unit: tCO2 / tCO2e; time: annual	derived; implemented; boundary=EGRID_NWPP	Meta annual MWh times eGRID subregion total output emission rate for ZIP 97754 → NWPP. Source/model: EPA_EGRID × Meta electricity. Limitation: Not hourly campus intensity.
Q_CI_PACW: PACW hourly consumed CO2 intensity	Unit: intensity units as prepared in pacw_hourly.csv; time: hourly	reported; implemented as regional shape; boundary=PACW_BA	EIA-930 consumed CO2 intensity for PACW where published (from 2018-07). Used as a regional relative hourly shape, not campus telemetry. Source/model: EIA930. Limitation: No consumed intensity before 2018-07. Not Meta-specific MEF.
Q_FUEL_IMPORT_PROXY: PACW fuel/import carbon score	Unit: index; time: hourly	proxy; implemented as sensitivity only; boundary=PACW_BA	Named sensitivity score from fuel mix and interchange when EIA consumed intensity is missing. Not the default shape when EIA intensity exists. Source/model: EIA930 fuel mix / interchange. Limitation: Must not be described as Meta-specific marginal emissions.
Q_MEF: Marginal emission factor	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented. No Meta-specific or PACW dispatch-marginal model. Source/model: . Limitation: Average location factors and a fuel/import score are not MEFs.
Q_BACKUP_GEN: Onsite backup generation hours	Unit: hours; time: monthly (gappy)	reported; implemented; boundary=META_ONSITE_BACKUP	DEQ annual-report monthly operating hours for permitted engines where extractable. Source/model: ODEQ_AIR_07_0037. Limitation: Exact hourly dispatch not available. 2019/2021 ARs scan-only.
Q_BACKUP_EMIS: Onsite backup emissions	Unit: tons as reported in DEQ tables (see processed files); time: monthly/annual as extracted	reported; implemented; boundary=META_ONSITE_BACKUP	DEQ-calculated annual-report emissions and separate source-test measurements. Source/model: ODEQ_AIR_07_0037. Limitation: Gaps in scan-only years; not grid carbon.
Q_OP_GHG: Operational Scope 1+2 (market-sensitive)	Unit: tCO2e; time: annual	reported; curated in annual table; not used as physical carbon model; boundary=META_CAMPUS	Meta-disclosed operational Scope 1+2 at the site table. Definition/source-vintage sensitive; 2018 REC agreement is an accounting breakpoint, not a physical switch. Source/model: Meta disclosures. Limitation: Not a location-based physical total.
Q_ELEC_COST: Electricity cost	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not identified. Source/model: . Limitation: No tariff or bill series.
Q_WATER_COST: Water cost	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not identified. Source/model: . Limitation: No water-rate series.
Q_CAPEX: Investment / capacity cost	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not identified. Permits do not state MW or sqft. Source/model: . Limitation: No cost or nameplate-capacity time series.
Q_WF: Water footprint (scarcity-weighted)	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented. Source/model: . Limitation: AWARE/Aqueduct factors are not in the pipeline.
Q_CF_SCARCITY: Scarcity characterization factor	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented. Source/model: . Limitation: IWA SUI is not a scarcity CF.
Q_WSF: Water-scarcity footprint	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented. Glossary: WSF is not a groundwater state. Source/model: . Limitation: Do not substitute IWA SUI for WSF or head.
Q_SPATIAL_M: Spatial crosswalk site → grid → generator → watershed → aquifer	Unit: ; time: static	derived; partial; boundary=META_CAMPUS	Partial: ZIP → eGRID NWPP and site point → HUC12 exist. Generator and aquifer maps for the campus are not identified. Source/model: EPA_EGRID_POWER_PROFILER; USGS_WBD. Limitation: No site → generator or site → aquifer network.
Q_DC_GW: Data-center groundwater extraction	Unit: m ³ ; time: unavailable as campus groundwater	unavailable; unavailable as the glossary quantity; boundary=NOT_IDENTIFIED	Groundwater-supplied share of campus withdrawal. Not identified. Direct POD exists but is not equated to campus withdrawal. Source/model: . Limitation: Needs source-share and/or well meters on the campus boundary.
Q_OTHER_SECTOR: Other-sector groundwater pumping	Unit: unavailable as wells; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented as a pumping series. USGS public-supply/irrigation are HUC12 models, not well pumping. Source/model: . Limitation: HUC12 USGS is not a well network.
Q_RECHARGE: Groundwater recharge	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not identified. ASR PDFs are context only. Source/model: . Limitation: No recharge series or model.
Q_HEAD: Groundwater head	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not identified. Do not substitute IWA surface availability. Source/model: . Limitation: Well-level GWIS BLS/AMSL observations exist as Q_GW_OBS. A modeled or spatially continuous hydraulic-head field is not identified. Mixed NGVD1929/NAVD1988 datums are not converted. Do not substitute IWA availability.
Q_STORAGE: Groundwater storage	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not identified. Source/model: . Limitation: No aquifer storage model.
Q_CONDUCTANCE: Inter-node conductance	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Groundwater network not identified. Source/model: . Limitation: No reduced-order or MODFLOW coupling.
Q_GW_OBS: Groundwater observations	Unit: ft below land surface; ft AMSL where GWIS reports it; time: irregular observation datetime	measured; implemented as measured GWIS well-level series; boundary=GWIS_WELL_OBSERVATIONS	Time-indexed OWRD GWIS measured water levels (ft below land surface) at matched and unmatched wells in the local export. GWIS-reported water-surface elevation is preserved with the site datum and is not recomputed from BLS+LSD. BLS and AMSL are paired representations of the same measurement, not independent observations. Not a fitted groundwater-head field. Source/model: OWRD_GWIS. Limitation: Numeric well levels are identified. Unresolved items are T/S/Sy/pumping-test parameters, combined Airport pumping, unmatched GWIS wells, and mixed datums — not missing GWIS BLS for the current export.
Q_SITING: Siting / location choice	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented. Out of current pipeline scope. Source/model: . Limitation: PSCC siting model not in this repository pipeline.
Q_EQUITY: Equity / burden metrics	Unit: unavailable; time: unavailable	unavailable; unavailable; boundary=NOT_IDENTIFIED	Not implemented. Source/model: . Limitation: Burden object not defined in code.

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Quantity	Unit / resolution	Provenance / status / boundary	Definition / source / limitation
Q_XI: Structural uncertainty terms	Unit: varies; time: ensemble	scenario; partial (scenario ensemble, not ξ identification); boundary=SCENARIO_WORKLOAD	Stochastic module samples facility-overhead and water-shape mixture priors; this is scenario uncertainty, not an identified structural-error model. Source/model: src/stochastic_conditional_simulation.py. Limitation: Do not treat ensemble bands as calibrated coverage probabilities.
Q_HEAT_RATE: Generator heat rate	Unit: mmBtu (CAMPD heat input) where present; time: hourly CAMPD	measured; partial (raw heat input present; not a campus HR model); boundary=OREGON_GENERATOR	CAMPD heat input is stored; a campus-coupled heat-rate model is not implemented. Source/model: EPA_CAMPD_OR. Limitation: No site-attributed heat rate.

Table 9: Machine-generated method/model registry (26 entries). Not every registered method is predictive.

Method / model	Class	Period / purpose	Selection / status / caveat
M_GRAYBOX: Prineville gray-box air-side physics	physics/accounting; prediction=no	Training: none (not statistically trained); holdout: n/a. Maps a supplied IT-power trace and hourly weather to facility power, PUE, cooling mode, and raw evaporative water. Does not infer IT power from annual totals.	Rule/status: n/a. Scaffold, not a digital twin. psychrolib used when installed; otherwise Buck + ASHRAE-form enthalpy. PRN1 late-2023/early-2024 permit evidence is not used to change these priors or to add a cooling break-point.
M_ELEC_CLOSURE: Annual facility-electricity closure (latent IT scale)	reconstruction; prediction=no — exact annual closure by construction	Training: all years 2011-2024 are closed (no electricity holdout); holdout: none for electricity. For each year, choose one constant IT-power scale so the gray-box annual facility energy equals Meta-reported MWh. This is calibration closure, not electricity prediction.	Rule/status: n/a. Never report the -0 residual as predictive accuracy.
M_WATER_SCALE_GLOBAL: Conditional water: global log-scale on raw evaporation	estimation; prediction=yes for annual water; in-sample for train	Training: through 2022 (years with positive raw and observed water); holdout: 2023-2024. Fit one multiplicative scale s on training years with observed withdrawal via geometric mean of $W_{\text{obs}}/W_{\text{raw}}$. Apply frozen s to all years including holdout. The scale is an empirical mapping from simplified raw-evaporation physics to the broader withdrawal accounting boundary, not a physical cooling multiplier.	Rule/status: BIC vs one-break; require BIC improvement > 2 to prefer one-break. Current selection: global, BIC=-11.43. Selected over the one-break alternative. Holdout 2023 +126.7%, 2024 +100.8% (MAPE 113.8%) is a two-year predictive diagnostic failure, not a formal statistical proof. Frozen training-mean baseline is not in this selection and currently performs much better on the two holdout years.
M_WATER_SCALE_ONEBREAK: Conditional water: one-break log-scale (candidate, not selected)	estimation; prediction=n/a (not selected)	Training: through 2022; holdout: 2023-2024 (unused because not selected). Optional two-scale model with a break year; 3 effective parameters (two scales + break). Not selected on current training data.	Rule/status: BIC; min_segment_years=3; require modest evidence over global. A selected break would still be statistical only, not a physical technology-change claim.
M_WATER_ENERGY_NULL: Annual water selected mechanistic candidate: energy-only	prediction; prediction=yes	Training: 2014-2022; holdout: 2023-2024. No-intercept nonnegative regression of annual withdrawal on reported facility MWh. Selected by expanding-window one-step MAPE among pre-registered mechanistic/covariate candidates. Not a claim that it is the best predictor overall; frozen naive baselines are outside that selection set.	Rule/status: expanding-window one-step MAPE among pre-registered mechanistic/covariate candidates; ties broken by fewer coefficients then name. Naive baselines are not in this set.. Selected mechanistic candidate among pre-registered energy/evaporation candidates (not a claim of best overall predictor). Expanding-window train MAPE 34.25%; frozen-NNLS holdout MAPE 135.2%. Frozen training-mean baseline holdout MAPE 24.5% was not in selection. Holdout N=2 is a diagnostic, not a formal proof.
M_WATER_EVAP_PHYS: Annual water candidate: evaporation-only	prediction; prediction=candidate only	Training: 2014-2022; holdout: 2023-2024 (not used for this unused model). No-intercept nonnegative regression of annual withdrawal on ensemble-median raw evaporation.	Rule/status: same expanding-window MAPE; score 52.06% (worse than energy_null). Not selected. Do not present as the pipeline's water forecast.
M_WATER_TWOCOMP: Annual water candidate: nonnegative energy + evaporation	prediction; prediction=candidate only	Training: 2014-2022; holdout: 2023-2024. NNLS on E_fac and raw evaporation. Current fit zeros the evaporation coefficient, collapsing to energy_null.	Rule/status: expanding-window one-step MAPE. NNLS dropped evaporation; still not selected because rolling MAPE is worse than the one-coefficient energy model.
M_STOCHASTIC: Stochastic conditional proxy (mixed Cox)	generative simulation; prediction=only the selected annual water model is a prediction; the rest is scenario/closure	Training: water model train through 2022; electricity/Scope 2 closed every year; holdout: 2023-2024 for predictive water only. Generative scenario: Cox-process arrivals, AR log-intensity, Poisson counts, Gamma work sizes, aggregate queue, utilization $\rightarrow$ IT shape, annual facility-energy scaling, sampled overhead priors, retrospective water-shape mixture, optional PACW carbon shape. Not recovered telemetry and not a general forecasting model.	Rule/status: water candidate selection as M_WATER_ENERGY_NULL. Default carbon allocates annual location Scope 2 in proportion to facility energy unless -use-pacw-shape. Stochastic water/PUE intervals are heuristic scenario ensembles under assumed workload/overhead priors, not calibrated confidence intervals with demonstrated coverage.
M_EGRID_BENCH: eGRID NWPP Meta MWh physical benchmark	benchmark; prediction=no	Training: n/a; holdout: n/a. Methodology/accounting consistency benchmark: Meta campus MWh times eGRID NWPP output rate versus Meta reported location Scope 2. Not electricity prediction, not a Meta-specific marginal model, and not fully independent external validation (both sides use the same Meta MWh and related inventory geography).	Rule/status: ZIP uniquely NWPP. Compare to Meta location Scope 2 where disclosed. 2011 eGRID tonnes are labeled eGRID_location_based_accounting_proxy because Meta Scope 2 is not separately disclosed. This is an accounting-method consistency check, not independent campus carbon telemetry. Market/REC not used.

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Method / model	Class	Period / purpose	Selection / status / caveat
M_PACW_CI: PACW EIA consumed-CO2 relative hourly shape	reconstruction; prediction=no	Training: n/a; holdout: n/a. Optional relative within-year carbon shape from PACW consumed intensity. Annual Meta location Scope 2 is still closed. Not campus telemetry.	Rule/status: prefer EIA consumed intensity when present; else fuel/import proxy. Coverage from 2018-07 for consumed intensity; demand from 2015-07.
M_FERC714_BACKCAST: constrained PACW-West proxy	FERC-hourly reconstruction; prediction=no	Training: none (not fit to EIA-930); holdout: 2016-2018 EIA overlap is independent validation; 2015 partial is secondary. Within each month, uses East+West hourly demand only for shape and West monthly NEL/peak/minimum for level. Affine $D_{hat_W} = mean_W + b_m (D_EW - mean_EW)$, $b_m \geq 0$ from peak/min, exact monthly energy closure to NEL_W. EIA-930 is not used to fit. Not reported PACW hourly demand and not campus electricity.	Rule/status: single nonnegative monthly scale; no second ML or overlap-calibrated backcast. Does not manufacture consumed CO2, fuel mix, hourly interchange, campus electricity, or marginal emissions. Does not overwrite pacw_hourly.csv.
M_FUEL_IMPORT: PACW fuel/import carbon score	benchmark; prediction=no	Training: n/a; holdout: n/a. Named sensitivity proxy only. Not default when EIA consumed intensity exists. Not a Meta-specific MEF.	Rule/status: fallback only. pacw_carbon_shape_compare.csv records correlation vs EIA intensity where both exist.
M_CHANGEPOINT: Annual piecewise-linear SSE break ranking	change-point screening; prediction=no	Training: train-only ranking uses years ≤ 2022 ; holdout: exploratory table uses all years and must not be used for holdout model selection. Ranks candidate break years by relative SSE reduction of two linear trends vs one, for electricity, water, and water intensity. Statistical candidates only.	Rule/status: ranking only; not auto-applied to physics epochs. Must not be named as a physical technology change unless independent permit/engineering evidence supports it. Train-only water intensity/water SSE reduction peaks at 2020.
M_IWA_IDENTITY: IWA availability accounting identity	physics/accounting; prediction=no	Training: n/a; holdout: n/a. Checks availab = strflow - consum in USGS IWA files. Structural QA, not hydrologic skill.	Rule/status: n/a. Never call this independent hydrologic validation.
M_OWRD_EXTERNAL: OWRD external water-evidence comparison	external-consistency check; prediction=no	Training: n/a (not a fit); holdout: n/a. Aligns modeled monthly campus water proxy with City production and direct POD without using them as calibration targets or summing boundaries.	Rule/status: n/a. Do not compute City vs Meta prediction error. Electricity closure is rechecked as an accounting identity.
M_OR_GEN_QC: Oregon CAMPD/EIA generator QC	external-consistency check; prediction=no	Training: n/a; holdout: n/a. Join/coverage diagnostics between CAMPD and EIA-923; cooling-water coverage; unit conversions. Not campus attribution.	Rule/status: n/a. Does not infer which generators served Prineville.
M_GW_SCAFFOLD: Groundwater observation scaffold	physics/accounting; prediction=no	Training: n/a; holdout: n/a. Compiles well identities, accepted OWRD pumping groups, GWIS measured water levels, construction/aquifer fields, and hydrogeologic parameter citations. Runs measurement-semantic QC then an identifiability screen on the eligible subset (within-well head_anomaly_ft = -(BLS - mean BLS) vs pumping at exploratory lags 0/1/3/6 months). Does not estimate heads, conductance, storage, or hydraulic connectivity.	Rule/status: n/a. Feasibility class is recomputed from GWIS+pumping evidence (not hard-coded). Identifiability_conclusion (A-small-subsystem-possible / B-validation-only / C-not-identifiable) is appended by the audit and may differ from broad Class B. ESTIMATION_CANDIDATE means sufficient data to attempt a validated empirical response model, not identified dynamics. HUC12 is not an aquifer node. No reduced-order model is fitted.
M_EWIF_PARTIAL: Partial-coverage Oregon cooling EWIF	physics/accounting; prediction=no	Training: n/a; holdout: n/a. Generation-weighted withdrawal/consumption intensity over cooling-intensity-eligible plant-months with non-missing water. Optional regional-average indirect-water proxy = EWIF $\times$ Meta MWh.	Rule/status: cooling_intensity_eligible and generation_mwh>0 and water not missing. Not Meta generator attribution. scientifically_meaningful_grid_ewif is false by construction.
M_DOCUMENTARY_EVIDENCE: Documentary/regulatory evidence integration	external-consistency check; prediction=no	Training: n/a; holdout: n/a. Validates curated documentary seeds, verifies raw PDF SHA-256 when files exist, and writes identity/event/evidence tables. Does not OCR PDFs and does not retune electricity/water models.	Rule/status: n/a. PRN and CCO remain distinct. 120/220/180/437 MW are not campus load. Water-infrastructure facts are not campus water observations. Meta annual Prineville totals are unchanged; reporting-boundary status is unresolved_prn_vs_prn_plus_cco.
M_CITY_SERVICE_CLIMATOLOGY: City-service month-of-year climatology baseline	benchmark; prediction=yes (retrospective exploratory)	Training: expanding complete years before each test year; holdout: retrospective/exploratory chronological; City data were inspected in development. Expanding-window mean of each calendar month of city_metered_water_service_m3 using only earlier complete years.	Rule/status: predeclared baseline; not selected by error minimization among a large library. Forecastable ex-ante baseline: earlier complete years only. Not untouched holdout. Response is City service, not Meta total withdrawal. Primary ranking uses common support.
M_CITY_SERVICE_SEASONAL_PERSIST: City-service seasonal persistence baseline	benchmark; prediction=yes (retrospective exploratory)	Training: previous complete year only; holdout: retrospective/exploratory chronological. Same calendar month in the previous complete year.	Rule/status: predeclared baseline. Forecastable ex-ante baseline: previous complete year only. Strong seasonal benchmark. Common-support ranking is primary; do not rank on native-support n.
M_CITY_SERVICE_ELEC_SCALE: City-service annual-electricity scale	prediction; prediction=yes (retrospective exploratory)	Training: expanding complete years; holdout: retrospective/exploratory chronological. Train-only nonnegative scale from Meta annual facility MWh to City-service annual total, allocated with climatology month shares.	Rule/status: predeclared. Contemporaneous explanatory: uses realized target-year annual electricity as a year-level scale. Not strictly prospective. Not hourly campus load.
M_CITY_SERVICE_GRAYBOX_LEVEL: City-service gray-box evaporation scale	prediction; prediction=yes (retrospective exploratory)	Training: expanding complete years; holdout: retrospective/exploratory chronological. Single nonnegative monthly scale from gray-box raw evaporation to City-service volume, fit on prior years only.	Rule/status: predeclared. Contemporaneous explanatory: uses realized target-month weather via raw evaporation and the reconstruction IT scale closed to that year's electricity. Not an ex-ante forecast. Does not overwrite the frozen Meta-annual water holdout.

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Method / model	Class	Period / purpose	Selection / status / caveat
M_CITY_SERVICE_SCALE_EVAP: service electricity + evaporation	City-prediction; prediction=yes (retrospective exploratory)	Training: expanding complete years; holdout: retrospective/exploratory chronological. Nonnegative two-term monthly model: $a \times (E_{\text{fac}}, y/12) + b \times W_{\text{raw_evap}}$.	Rule/status: predeclared. Contemporaneous explanatory: uses target-year annual electricity and realized target-month weather. No ML, no hyperparameter search, no structural-break fitting. Not strictly prospective.
M_CITY_SERVICE_GRAYBOX_SHAPE: Gray-box vs City-service within-year monthly shares	Gray-external-consistency check; prediction=no	Training: n/a (within-year normalization); holdout: n/a. Compares $p_{\text{obs}} = W_{\text{city}}(m)/\text{sum}_m W_{\text{city}}$ to $p_{\text{gray}} = W_{\text{raw_evap}}(m)/\text{sum}_m W_{\text{raw_evap}}$ for complete years. Shape only; levels are not forced equal.	Rule/status: n/a. Asks whether evaporative physics has approximately the right seasonal mechanism independently of annual campus/source accounting.
M_CITY_BOUNDARY_RECONCILE: Meta annual vs City meter component reconciliation	Meta-external-consistency check; prediction=no	Training: n/a; holdout: n/a. Year-by-year residuals and shares of Meta annual withdrawal versus City-service and service+bulk (bulk allocated by observed bill year). Descriptive diagnostics only; not a fitted closure and not selection of a physical total.	Rule/status: none; minimizing residual does not identify the campus boundary. Shares and residuals are descriptive. 2022 City service $\approx$ Meta and 2023–2024 service+bulk (bill year) $\approx$ Meta are accounting-date diagnostics, not water-balance closure and not physical identification. City-service/Meta share is not historically stable.

C City water-boundary reconciliation diagnostic

This appendix aggregates only complete City-service years that overlap Meta’s reported annual withdrawal. Service + bulk is shown only where all 12 bulk billing months are present. Every combination remains diagnostic only—not an identified campus mass balance.

Table 10: Annual boundary-reconciliation diagnostic for complete City-service years. Service + bulk is diagnostic only – not an identified campus mass balance; blank bulk combinations lack 12 billing months.

Year	Meta	City service	Svc./Meta	Bulk	Service+bulk	(S+B)/Meta
2014	39,747	5,000	0.126			
2016	70,000	32,769	0.468			
2017	172,000	86,355	0.502			
2018	160,000	120,859	0.755			
2019	208,000	177,673	0.854	19,344	197,017	0.947
2020	445,000	328,339	0.738	43,067	371,407	0.835
2021	354,000	380,331	1.074	41,163	421,494	1.191
2022	240,000	239,368	0.997	26,493	265,862	1.108
2023	180,000	164,878	0.916	14,873	179,751	0.999
2024	328,000	320,966	0.979	5,484	326,450	0.995

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